

Security bulletins | Cloud Customer Care

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Content

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Security bulletins

The following security bulletins are related to Google Cloud products.

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GCP-2026-066

Published: 2026-09-28

Description

| Description | Severity | Notes | |

A Confused Deputy vulnerability was discovered in the Email Task component in Application Integration versions prior to June 30, 2026.

What should I do?

No customer action is required. This vulnerability was patched on June 30, 2026.

What vulnerabilities are being addressed?

An authenticated attacker could read and exfiltrate arbitrary Google-internal files using a crafted attachment path.

| High | [CVE-2026-81375](#) | |

GCP-2026-065

Published: 2026-09-28

Description

| Description | Severity | Notes | | |

A Deserialization of Untrusted Data vulnerability was discovered in the JavaScript Task in Application Integration versions prior to June 28, 2026.

What should I do?

No customer action is required. This vulnerability was patched on June 28, 2026.

What vulnerabilities are being addressed?

An authenticated user with standard permissions could use a specially crafted script to run arbitrary code on the shared production servers.

| Critical | [CVE-2026-81867](#) | |

GCP-2026-064

Published: 2026-09-28

Description

| Description | Severity | Notes | | |

An Incorrect Authorization vulnerability was discovered in the task configuration in Application Integration versions prior to June 17, 2026.

What should I do?

No customer action is required. This vulnerability was patched on June 17, 2026.

What vulnerabilities are being addressed?

An authenticated user could use an internal-only task type to execute arbitrary internal RPCs from the Google-internal production network under a privileged identity.

| Critical | [CVE-2026-19759](#) | |

GCP-2026-063

Published: 2026-09-16

Description

| Description | Severity | Notes | | |

An Improper Privilege Management vulnerability in Google Security Operations SOAR was patched with version 6.3.85, and no customer action is needed. The vulnerability allowed an authenticated attacker to escalate privileges to system-level administrative access using a crafted internal authentication header.

What should I do?

No customer action is required. All customers have been automatically upgraded to the fixed version: 6.3.85 or higher.

| Critical | [CVE-2026-15587](#) | |

GCP-2026-062

Published: 2026-09-11

Description

| Description | Severity | Notes | | |

Multiple security vulnerabilities were discovered in Slurm that affect Cluster Toolkit blueprints that reference specific image versions. These vulnerabilities affect the `slurmstepd` daemon, RPC request handling, and the accounting database.

What should I do?

For instructions and more details, see the [Cluster Toolkit security bulletin](#).

| High |

- [CVE-2026-65107](#)
- [CVE-2026-65108](#)
- [CVE-2026-65109](#)
- [CVE-2026-65138](#)
- [CVE-2026-65139](#)
- [CVE-2026-65140](#)
- [CVE-2026-65165](#)
- [CVE-2026-65168](#)

| |

GCP-2026-061

Published: 2026-09-09

Description

| Description | Severity | Notes | | |

A security vulnerability ([GHSA-p7v4-vr35-mj6f](#), CVE assignment pending) in containerd's CRI implementation allows a container restored from an untrusted checkpoint through the `CreateContainer` API to bypass the destination security context and execute with elevated privileges.

GKE clusters are not vulnerable by default. GKE node images do not include the `criu` tool.

What should I do?

No immediate customer action is required for default clusters. If you install custom runtime software or node tools on your nodes that include the `criu` binary, refer to the security bulletin for recommended actions.

What vulnerabilities are being addressed?

For more information, see the [GKE security bulletin](#).

| Medium | [GHSA-p7v4-vr35-mj6f](#) | |

GCP-2026-060

Published: 2026-09-04

Updated: 2026-09-08

Description

| Description | Severity | Notes | | |

2026-09-08 Update: Added link to the Cluster Toolkit security bulletin.

A security flaw in the Slurm `sbcast` tool (CVE-2026-65107) lets shared library files bypass security checks and can crash nodes in your cluster.

What should I do?

For instructions and more details, see the following bulletins:

- [Cluster Toolkit security bulletin](#)
- [Cluster Director security bulletin](#)

| High | [CVE-2026-65107](#) | |

GCP-2026-059

Published: 2026-09-04

Description

| Description | Severity | Notes | | |

A Missing Authorization vulnerability was discovered in the HTTP Connector in Integration Connectors versions prior to December 11, 2025.

What should I do?

No customer action is required. This vulnerability was patched on December 11, 2025.

What vulnerabilities are being addressed?

An authenticated attacker could use an unauthorized service account attachment to escalate privileges and take over a Google Cloud project.

| High | [CVE-2026-4644](#) | |

GCP-2026-058

Published: 2026-09-02

Description

| Description | Severity | Notes | | |

A missing project permission check in GKE Multi-Cloud (`CreateAttachedCluster`, `CreateAwsCluster`, `CreateAzureCluster`) APIs allowed an attacker to register an attached cluster into an arbitrary target project's Workload Identity Federation for GKE. This registration allowed unauthorized creation of Workload Identity tokens and impersonation of Kubernetes Service Accounts that have bindings in the target project through the Workload Identity Federation for GKE configuration.

For instructions and more details, see the [GKE security bulletin](#)

| Critical | |

GCP-2026-057

Published: 2026-08-26

Description

| Description | Severity | Notes | | |

A series of vulnerabilities were discovered in Envoy Proxy.

For instructions and more details, see the [Cloud Service Mesh security bulletin](#).

| Moderate to High |

- [CVE-2026-73513](#)
- [CVE-2026-73552](#)
- [CVE-2026-73512](#)
- [CVE-2026-73547](#)
- [CVE-2026-73549](#)

- [CVE-2026-50572](#)
- [CVE-2026-73546](#)
- [CVE-2026-48521](#)
- [CVE-2026-73551](#)
- [CVE-2026-73511](#)
- [CVE-2026-73548](#)
- [CVE-2026-73550](#)
- [CVE-2026-73553](#)

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GCP-2026-056

Published: 2026-08-26

Description

| Description | Severity | Notes | | |

An Improper Input Validation vulnerability was discovered in the [JDBC driver](#) in BigQuery Data Transfer Service versions prior to May 1, 2026.

What should I do?

No customer action is required. This vulnerability was patched on May 1, 2026.

What vulnerabilities are being addressed?

An authenticated attacker could use crafted JDBC connection string parameters to achieve remote code execution in the connector container and escalate privileges in the tenant project.

| Critical | [CVE-2026-12717](#) | |

GCP-2026-055

Published: 2026-08-25

Description

| Description | Severity | Notes | | |

A critical unauthenticated Remote Code Execution (RCE) vulnerability exists in Next.js and `libheif` when processing malicious image files. Google Cloud infrastructure is not directly vulnerable, but customer workloads running Next.js on Google Cloud (e.g., Cloud Run, GKE, App Engine) may be affected.

What should I do?

Google Cloud backend services are not directly impacted, and no action is required to secure the underlying Google Cloud infrastructure. However, customers hosting Next.js applications on Google Cloud must take immediate action to secure their own workloads. We recommend that you manually upgrade your Next.js deployments to one of the following versions (or later) in your `package.json`:

- 16.3.3
- 15.5.24

Customers utilizing `libheif` should monitor for upstream OS patches and rebuild their container base images accordingly. Redeploy your updated workloads to ensure running containers utilize the patched versions.

What vulnerabilities are being addressed?

The vulnerabilities, [GHSA-2xp9-vwfh-vxw4](#) and [GHSA-g89c-p67h-r497](#), allow an attacker to execute arbitrary code within the context of the application by supplying a crafted image file during Next.js image optimization. This issue occurs due to how the underlying `libheif` library processes clean aperture boxes in HEIF/AVIF images.

| High | [GHSA-2xp9-vwfh-vxw4](#) [GHSA-g89c-p67h-r497](#) | |

GCP-2026-054

Published: 2026-08-11

Description

| Description | Severity | Notes | |

A vulnerability (CVE-2026-6726, also known as TCGVRT0010) has been identified in the Trusted Computing Group's TPM 2.0 reference implementation code, affecting all published revisions of the code (v1.16, v1.38, v1.59, v1.83, v184).

What should I do?

No customer action is required. Google will proactively update your systems during your standard and planned maintenance windows.

What vulnerabilities are being addressed?

For more information, see the [Compute Engine security bulletin](#).

| High | [CVE-2026-6726](#) | |

GCP-2026-053

Published: 2026-08-11

Description

| Description | Severity | Notes | |

Google is aware of several security vulnerabilities affecting Intel® Trust Domain Extensions (TDX) firmware that can compromise confidential guest integrity. These vulnerabilities could allow a privileged host adversary to bypass attestation checks, access restricted registers, or decrypt normally protected guest memory.

What Google products are affected?

All Confidential VM instances relying on Intel TDX technology.

What should I do?

Google has proactively applied these firmware upgrades to our server fleet to mitigate these vulnerabilities. No action is required for customers that did not receive a separate direct advisory to upgrade.

For more information, see Intel's PSIRT technical advisories: [INTEL-TA-01404](#), [INTEL-TA-01428](#), [INTEL-TA-01419](#), [INTEL-TA-01439](#), [INTEL-TA-01442](#), [INTEL-TA-01436](#).

| High |

- [CVE-2025-31938](#)
- [CVE-2025-35973](#)
- [CVE-2025-31356](#)
- [CVE-2026-20898](#)
- [CVE-2026-20713](#)
- [CVE-2026-20901](#)
- [CVE-2026-20885](#)
- [CVE-2026-20705](#)
- [CVE-2026-20775](#)

| |

GCP-2026-052

Published: 2026-08-07

Description

| Description | Severity | Notes | |

A security vulnerability has been identified in the Kernel-based Virtual Machine (KVM) x86 shadow Memory Management Unit (MMU) impacting both Intel and AMD platforms.

What should I do?

No action is required for managed Compute Engine virtual machines. Because the vulnerability resides in the host hypervisor, Google has mitigated all affected Intel and AMD-based host infrastructure transparently via live patching without requiring customer VM downtime, reboots, or manual upgrades.

If you operate self-managed virtualized environments or custom host hypervisors outside standard managed Compute Engine infrastructure, you should ensure your host Linux kernel is updated with the upstream patch as soon as it is made available by your operating system distribution vendor.

What vulnerabilities are being addressed?

This is a KVM vulnerability that could lead to a VM escape. There has been no active exploitation observed, and this is a proactive patching effort.

| None | [CVE-2026-64561](#) | |

GCP-2026-051

Published: 2026-07-31

Description

2026-08-04 Update: The VMware patch release version is 1.33.1200, not 1.33.1100 as stated in the initial bulletin.

| Description | Severity | |

The following vulnerabilities have been discovered in containerd (the GDC container runtime). These vulnerabilities allow attackers with permissions to create Pods to bypass Kubernetes security boundaries and perform host compromise, cache poisoning, and denial of service. While these vulnerabilities are critical in the context of containerd, the requirement to have cluster privileges to create Pods to exploit them means they are considered High according to GDC (software only) vulnerability classification.

- **CVE-2026-50195 (Critical):** containerd's CRI checkpoint import process fails to validate image references. An attacker with permissions to create Pods can use a crafted checkpoint image to poison the node's local image cache, causing other pods to use a malicious image.
- **CVE-2026-53488 (Critical):** CRI plugin propagates labels from an image config to a container without validation. This may result in executing an arbitrary command on the host.
- **CVE-2026-53492 (Critical):** CRI implementation improperly trusts Container Device Interface (CDI) annotations during container restoration. This allows an attacker to inject arbitrary CDI configurations (such as host mounts and device nodes) into restored containers, bypassing resource allocation and device plugin enforcement.
- **CVE-2026-53489 (High):** CRI plugin restores container.log without validating a symlinked path, allowing an attacker to read arbitrary host files via kubectl logs.
- **CVE-2026-47262 (Moderate):** A vulnerability in containerd allows a maliciously crafted image to cause memory exhaustion (DoS) of the containerd process.

These vulnerabilities affect all GDC (software only) configurations using Ubuntu node images.

What should I do?

For GDC software only on bare metal, the following patches will contain fixes for the vulnerabilities issue:

- 1.34.800
- 1.33.1100

For GDC software only on VMware, the following patches will contain fixes for the vulnerabilities:

- 1.34.800
- 1.33.1200

| High | |

GCP-2026-050

Published: 2026-07-29

Description

| Description | Severity | Notes | | |

Per advisory VMSA-2026-0006, multiple vulnerabilities in VMware ESXi, and vCenter were privately reported to Broadcom. We are in the process of applying the necessary patches supplied by Broadcom. There are no known workarounds for these reported vulnerabilities.

Once patched, your VMware Engine deployments should be running vCenter and ESXi 8.0 U3k.

What should I do?

Google recommends customers to monitor their workloads on VMware Engine for any unusual activities.

| Critical |

- [VMSA-2026-0006](#)
- CVE-2026-59309
- CVE-2026-59310
- CVE-2026-47876
- CVE-2026-41703
- CVE-2026-41709

| |

GCP-2026-049

Published: 2026-07-22

Description

| Description | Severity | Notes | |

A reflected Cross-Site Scripting (XSS) vulnerability was discovered in Google Cloud Looker. An attacker could craft a malicious URL that, when opened by an authenticated administrator, would allow the attacker to execute arbitrary scripts on their behalf and potentially compromise the administrator account.

What should I do?

For Looker-hosted instances, no action is required.

For self-hosted Looker instances, update your Looker instances as soon as possible. This vulnerability has been patched in all supported versions of Looker for self-hosted instances. The following versions have all been updated to fix this vulnerability:

- Looker 26.8.7 and all later versions
- Looker 26.6.28+
- Looker 26.4.36+
- Looker 26.2.47+
- Looker 26.0.66+
- Looker 25.18.68+
- Looker 25.12.65+
- Looker 25.6.103+

What vulnerabilities are being addressed?

For more information, see [CVE-2026-15810](#).

| High | [CVE-2026-15810](#) | |

GCP-2026-048

Published: 2026-07-13

Description

| Description | Severity | Notes | |

A privilege escalation vulnerability was addressed in Developer Connect. Previously, for GitLab Enterprise and Bitbucket Data Center connections, when Secret Manager secrets were retrieved, permissions were checked against the Developer Connect service agent (P4SA) credentials only. Developer Connect now validates that both the calling principal and the P4SA have the required permissions on the referenced secrets.

For instructions and more details, see the [Developer Connect security bulletin](#).

| Medium | |

GCP-2026-047

Published: 2026-07-13

Description

| Description | Severity | Notes | |

A Missing Authorization vulnerability was discovered in repositories in BigQuery, Dataform, and Colab Enterprise.

What should I do?

No customer action is required. Google has already applied mitigations to all impacted products and services.

What vulnerabilities are being addressed?

During repository creation, an authenticated attacker could potentially escalate their permissions and perform cross-tenant repository takeover.

| Critical | [CVE-2026-14934](#) | |

GCP-2026-046

Published: 2026-07-06

Description

| Description | Severity | Notes | |

A virtualization vulnerability has been identified in the KVM x86 shadow paging and nested virtualization configurations. Because an attacker operating a virtual machine with nested virtualization could escape hypervisor boundary isolation to compromise the underlying physical host, any x86 virtual machine (whether nested or non-nested) hosted on an Intel-based server supporting nested virtualization is potentially exposed.

What should I do?

Google is deploying live hypervisor hotpatches across all managed Compute Engine host servers globally. No action is required for managed Compute Engine virtual machines or Google Kubernetes Engine clusters. Because remediation is performed transparently at the physical host hypervisor level, customer VM downtime, reboots, or manual upgrades are not required.

Customers operating self-managed virtualized environments or custom host hypervisors outside standard managed Compute Engine infrastructure should ensure their host Linux kernel is updated with the upstream patch as soon as it is made available by their operating system vendor.

What vulnerabilities are being addressed?

The vulnerability (CVE-2026-53359) exploits a use-after-free error in the x86 shadow paging and nested page table translation routines. Google has observed no evidence of active customer exploitation in production environments. Live patching and host-level monitoring have been implemented globally to mitigate risk and detect exploit attempts.

| High/S0 | [CVE-2026-53359](#) (Januscape) | |

GCP-2026-045

Published: 2026-06-29

Description

| Description | Severity | Notes | | |

A vulnerability was detected in Envoy Proxy where blocked QPACK decoding can cause a Denial-of-Service Attack against the HTTP/3 stack.

For instructions and more details, see the [Cloud Service Mesh security bulletin](#).

| High | [GHSA-p7c7-7c47-pwch](#) | |

GCP-2026-044

Published: 2026-06-25

Description

| Description | Severity | Notes | | |

A vulnerability was detected in Application Integration's JavaScript task, which was using the Rhino JavaScript engine. This only impacted tasks published before January 2025.

For details and instructions, see the [Application Integration security bulletin](#).

| High | [CVE-2025-0982](#) | |

GCP-2026-043

Published: 2026-06-24

Description

| Description | Severity | Notes | | |

A vulnerability was found in Firebase Studio where the `GetSignedGcsUrl` RPC allowed authenticated users to list buckets and download deployment source code of other tenants.

No action is required to mitigate this vulnerability as the fix has been deployed to the backend service.

As a precautionary measure, users who stored sensitive information, such as API keys (for example, `GEMINI_API_KEY`), within their Firebase Studio workspace may choose to rotate these keys. For instructions, see the [Firebase Studio troubleshooting guide](#).

| High | [CVE-2026-12715](#) | |

GCP-2026-042

Published: 2026-06-24

Description

| Description | Severity | Notes | | |

A privilege escalation vulnerability was addressed in Cloud Build. Previously, for GitLab Enterprise and Bitbucket Data Center connections, when Secret Manager secrets were retrieved, permissions were checked against the Cloud Build service agent (P4SA) credentials only. Cloud Build now validates that both the calling principal and the P4SA have the required permissions on the referenced secrets.

For instructions and more details, see the [Cloud Build security bulletin](#).

| Low | | |

GCP-2026-041

Published: 2026-06-24

Description

| Description | Severity | Notes | | |

A recent security investigation was conducted regarding log4j vulnerabilities (CVE-2026-34480 and CVE-2026-34477) reported in Apigee Edge for Private Cloud. Google has determined that Apigee Edge for Private Cloud is **not vulnerable** to these CVEs. **No customer action is required.**

For more information, see the [Apigee security bulletin](#).

| High | [CVE-2026-34480](#) [CVE-2026-34477](#) | |

GCP-2026-040

Published: 2026-06-23

Description

| Description | Severity | Notes | | |

A series of vulnerabilities were discovered in Envoy Proxy.

For instructions and more details, see the [Cloud Service Mesh security bulletin](#).

| Moderate to High | [CVE-2026-47692](#) [CVE-2026-47207](#) [CVE-2026-47205](#) [CVE-2026-47220](#) [CVE-2026-47221](#) [CVE-2026-48044](#) [CVE-2026-48090](#) [CVE-2026-47778](#) [CVE-2026-47204](#) [CVE-2026-48497](#) [CVE-2026-48706](#) [CVE-2026-48743](#) [CVE-2026-47775](#) [CVE-2026-48042](#) | |

GCP-2026-039

****Published:** **2026-06-22

Description

| Description | Severity | Notes | | |

A vulnerability in Cloud Logging could have allowed attackers to hijack log sink destinations by recreating the target Cloud Storage bucket in a project controlled by the attacker. This issue could have resulted in the continuous routing of sensitive logs to an unauthorized third party.

For more information, see the [Google Cloud Observability security bulletin](#).

| Medium | |

GCP-2026-038

****Published:** **2026-06-22

Description

| Description | Severity | Notes | | |

A vulnerability was found in App Engine where the `GetDashboardAppStats` GraphQL private API operation in the Google Cloud console leaked cross-tenant request logs.

No action is required. This vulnerability has been fixed.

| High | [CVE-2026-8934](#) | |

GCP-2026-037

Published: 2026-06-18

Updated: 2026-06-20

Description

| Description | Severity | Notes | | |

2026-06-20 Update: Added GKE patch versions containing the fixes for Container-Optimized OS node images for minor versions 1.35 and 1.36. Patches for Ubuntu node images are still pending and in progress.

2026-06-19 Update: Added GKE patch versions containing the fixes for Container-Optimized OS node images for minor versions from 1.30 to 1.34. Patches for Ubuntu node images and for minor version 1.35 and 1.36 for Container-Optimized OS node images are in progress.

Multiple vulnerabilities have been discovered in containerd (the GKE container runtime). These vulnerabilities allow attackers with permissions to create pods to bypass Kubernetes security boundaries and perform host compromise, cache poisoning, and denial of service.

For instructions and more details, see the [GKE security bulletin](#)

| Critical | [CVE-2026-50195](#) [CVE-2026-53488](#) [CVE-2026-53492](#) [CVE-2026-53489](#) [CVE-2026-47262](#) | |

GCP-2026-036

****Published:** **2026-06-09

Description

| Description | Severity | Notes | | |

ARM has announced CVE-2025-10263, an architectural issue affecting some Arm cores which lets an attacker bypass translation stages or GPT protections under certain conditions. This vulnerability lets an attacker at a lower exception level to write to memory owned by a higher exception level, thereby escalating privileges. This issue does not affect memory reads.

For more information, see the following bulletins:

- [Compute Engine security bulletin](#)

| High | [CVE-2025-10263](#) | |

GCP-2026-035

Published: 2026-06-08

Description

| Description | Severity | Notes | | |

A vulnerability was found in Envoy where HTTP/2 downstream request processing allow an unauthenticated remote client to trigger excessive memory consumption, potentially resulting in OOM termination of the Envoy process and denial of service.

For instructions and more details, see the [Cloud Service Mesh security bulletin](#).

| High | [CVE-2026-47774](#) | |

GCP-2026-034

****Published: ****2026-05-20

Description

| Description | Severity | Notes | |

A vulnerability was found in Apigee where the `IntegrationRegion` parameter in the `SetIntegrationRequest` policy lacks validation, allowing for Server-Side Request Forgery (SSRF) and service account token exfiltration. The issue arises when an attacker can control a flow variable used for `IntegrationRegion`, leading to requests being sent to an attacker-controlled host with the service account token.

| High | [CVE-2026-2264](#) | |

GCP-2026-033

****Published: ****2026-05-14

****Updated: ****2026-07-16

Description

| Description | Severity | Notes | |

2026-07-16 Update: The following versions of GKE are updated with code to fix this vulnerability on Ubuntu. Upgrade your Ubuntu node pools to the following versions or later:

- 1.30.14-gke.2710000
- 1.31.14-gke.2116000
- 1.32.13-gke.1829000
- 1.33.13-gke.1011000
- 1.34.9-gke.1131000
- 1.35.6-gke.1127000
- 1.36.2-gke.1346000

A container breakout vulnerability CVE-2026-46300 has been found in the Linux kernel, known as Fragnesia. It allows an unprivileged local attacker to escalate to root on the host.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| Medium | [CVE-2026-46300](#) | |

GCP-2026-032

****Published:** **2026-05-12

Description

| Description | Severity | Notes | | |

AMD has identified a hardware-level vulnerability in **Zen 2 microarchitecture processors** (including EPYC and Ryzen series) involving potential corruption within the **micro-operation (OP) cache**. Under specific conditions, this issue (AMD-SN-7052 / CVE-2025-54518) could lead to security boundary bypasses or unauthorized data access. We have deployed fixes across Google infrastructure to mitigate these issues.

For more information, see the [Compute Engine security bulletin](#).

| High | [CVE-2025-54518](#) | |

GCP-2026-031

****Published:** **2026-05-12

Description

| Description | Severity | Notes | | |

Researchers discovered a vulnerability in AMD firmware that, due to missing protection, could allow a malicious hypervisor to execute arbitrary code on the AMD Secure Processor (ASP). This allows for the escalation of Memory Mapped I/O (MMIO) read and write permissions, which compromises the confidentiality and integrity of SEV-SNP guests.

For more information, see the [Compute Engine security bulletin](#).

| Medium |

- [CVE-2025-61971](#)
- [CVE-2025-61972](#)
- [CVE-2024-36315](#)

| |

GCP-2026-030

****Published:** **2026-05-11

Updated: 2026-06-24

Description

| Description | Severity | Notes | | |

2026-06-24 Update: Added patch versions for GKE.

2026-05-20 Update: Added CVE-2026-43500 and added CVE IDs to exploit paths.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- CVE-2026-43284
- CVE-2026-43500

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2026-43284](#) | |

GCP-2026-029

****Published:** **2026-05-07

Description

| Description | Severity | Notes | |

Microsoft is updating the Secure Boot certificates originally issued in 2011 to ensure Windows devices continue to verify trusted boot software. These older certificates begin expiring in June 2026. Devices that haven't received the newer 2023 certificates will continue to start and operate normally, and standard Windows updates will continue to install. However, these devices will no longer be able to receive new security protections for the early boot process, including updates to Windows Boot Manager, Secure Boot databases, revocation lists, or mitigations for newly discovered boot-level vulnerabilities. Also, Secure Boot certificate expirations starting in June 2026 affect Linux systems that use Secure Boot.

What should I do?

Google recommends that customers update their Windows VMs by taking appropriate actions as recommended by Microsoft. Distributions like Ubuntu, Red Hat, and Fedora are already working to provide updated packages signed with the new 2023 key. Refer also to the [Broadcom documentation](#) to resolve errors and warnings in VMware virtual machines as Secure Boot certificates approach expiration. After June 2026, systems lacking the 2023 certificate updates may experience failures during new operating system installations or while updating the existing bootloader firmware.

| Informational | [Broadcom KB 423893](#) | |

GCP-2026-028

****Published:** **2026-05-05

Updated: 2026-05-27

Description

| Description | Severity | Notes | | |

[CVE-2026-31431](#), also known as "Copy Fail," is a high-severity local privilege escalation (LPE) vulnerability in the Linux kernel that allows an unprivileged user to gain root access. Disclosed in late April 2026, it stems from a logic flaw in the kernel's cryptographic subsystem (algif_aead) introduced in 2017.

What should I do?

Google recommends that customers protect their Linux Guest VMs by updating the kernel on all Linux VMs. Major distributions have released or are rolling out fixes.

| High | [CVE-2026-31431](#) | |

GCP-2026-027

****Published:** **2026-05-01

Description

| Description | Severity | Notes | | |

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- [CVE-2026-23351](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2026-23351](#) | |

GCP-2026-026

****Published:** **2026-04-30

Updated: 2026-05-04

Description

| Description | Severity | Notes | | |

2026-05-04 Update: Added patch versions for GKE.

A vulnerability in the Linux kernel ([CVE-2026-31431](#)) allows an unprivileged local attacker to write to the system page cache, potentially leading to local privilege escalation and container escape.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2026-31431](#) | |

GCP-2026-025

****Published: ****2026-04-30

Description

| Description | Severity | Notes | | |

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2026-23274

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2026-23274](#) | |

GCP-2026-024

****Published: ****2026-04-28

Description

| Description | Severity | Notes | | |

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- [CVE-2025-38248](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-38248](#) | |

GCP-2026-023

****Published:** **2026-04-28

Updated: 2026-05-07

Description

| Description | Severity | Notes | |

2026-05-07 Update: Added patch versions for Ubuntu node pools on GKE

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- [CVE-2026-23074](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2026-23074](#) | |

GCP-2026-022

****Published:** **2026-04-16

Description

| Description | Severity | Notes | |

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- [CVE-2026-23209](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2026-23209](#) | |

GCP-2026-021

Published: 2026-04-14

Description

| Description | Severity | Notes | | |

AMD reported a vulnerability in its firmware that could have allowed a malicious hypervisor to direct the IOMMU to write into the guest memory of AMD SEV-SNP enabled instances, compromising guest data integrity. Google rolled out a mitigation to vulnerable Confidential VM instances with AMD SEV-SNP enabled.

What should I do?

No customer action is needed. The mitigation has already been applied to Confidential VM instances with AMD SEV-SNP enabled.

For more information, see AMD advisory [AMD-SB-3016](#).

| Medium |

[CVE-2023-20585](#)

| |

GCP-2026-020

****Published:** **2026-04-14

Description

| Description | Severity | Notes | | |

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- [CVE-2026-23231](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2026-23231](#) | |

GCP-2026-019

****Published:** **2026-04-14

Description

| Description | Severity | Notes | | |

Researchers discovered a vulnerability in AMD firmware that could allow a malicious hypervisor to alter BIOS settings and Memory Mapped I/O (MMIO) routing configurations, compromising the confidentiality and integrity of Confidential VMs with AMD SEV-SNP guests.

For more information, see the [Compute Engine security bulletin](#).

| Medium | [CVE-2025-54510](#) | |

GCP-2026-018

****Published:** **2026-04-07

Updated: 2026-04-15

Description

| Description | Severity | Notes | | |

2026-04-15 Update: Added patch versions for Ubuntu nodes with GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2026-23111

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)

- [GDC software for bare metal security bulletin](#)

| High | [CVE-2026-23111](#) | |

GCP-2026-017

****Published:** **2026-04-02

Description

| Description | Severity | Notes | | |

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2026-23273

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2026-23273](#) | |

GCP-2026-016

****Published:** **2026-03-27

Updated: 2026-03-31

Description

| Description | Severity | Notes | | |

2026-03-31 Update: Added patch versions for Ubuntu nodes with GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2025-38616

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-38616](#) | |

GCP-2026-015

Published: 2026-03-27

Description

| Description | Severity | Notes | | |

The following vulnerability was discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes: [CVE-2026-23268](#) .

For more information, see the [Compute Engine security bulletin](#):

| High | [CVE-2026-23268](#) | |

GCP-2026-014

****Published:** **2026-03-14

Updated: 2026-03-25

Description

| Description | Severity | Notes | | |

2026-03-25 Update: Added patch versions for Ubuntu nodes with GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- [CVE-2025-38678](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-38678](#) | |

GCP-2026-013

Published: 2026-03-11

Description

| Description | Severity | Notes | |

The following vulnerabilities were discovered in Envoy Proxy:

- CVE-2026-26311
- CVE-2026-26309
- CVE-2026-26310
- CVE-2026-26308
- CVE-2026-26330

The following vulnerabilities were discovered in Istio:

- CVE-2026-31837
- CVE-2026-31838

For instructions and more details, see the [Cloud Service Mesh security bulletin](#).

| Medium to High |

- [CVE-2026-26311](#)
- [CVE-2026-26309](#)
- [CVE-2026-26310](#)
- [CVE-2026-26308](#)
- [CVE-2026-26330](#)
- [CVE-2026-31837](#)
- [CVE-2026-31838](#)

| |

GCP-2026-012

Published: 2026-02-20

Description

| Description | Severity | Notes | |

In Google Cloud Vertex AI, a vulnerability involving predictable bucket naming was identified in Vertex AI Experiments from version 1.21.0 up to (but not including) 1.133.0.

What should I do?

No customer action is needed for mitigation.

CVE-2026-2473 allows an unauthenticated remote attacker to achieve cross-tenant remote code execution, model theft, and poisoning using pre-creating predictably named Cloud Storage buckets (Bucket Squatting). This vulnerability was identified in Vertex AI Experiments version 1.21.0. Mitigations have already been applied to version 1.133.0 and later.

| High | [CVE-2026-2473](#) | |

GCP-2026-011

Published: 2026-02-20

Description

| Description | Severity | Notes | |

A stored Cross-site Scripting (XSS) vulnerability in `_genai/_evals_visualization` was identified in Google `google-cloud-aiplatform` (Vertex AI Python SDK Visualization) on Exclusively-Hosted-Service.

What should I do?

Customers will need to update their `google-cloud-aiplatform` Python SDK to version 1.131.0 (released on 2025-12-16) or later to receive the fix.

CVE-2026-2472 allows an unauthenticated remote attacker to execute arbitrary JavaScript in a victim's Jupyter or Colab environment using injecting script escape sequences into model evaluation results or dataset JSON data. This vulnerability was identified in Google `google-cloud-aiplatform` (Vertex AI Python SDK) prior to 1.131.0.

| High | [CVE-2026-2472](#) | |

GCP-2026-010

Published: 2026-02-13

Description

| Description | Severity | Notes | |

A vulnerability was identified in the Apigee platform that could have allowed a malicious actor with administrative or developer-level permissions in their own Apigee environment to elevate privileges and access cross-tenant data.

Specifically, a vulnerability in Apigee's sandboxing technology allowed the use of a link-local endpoint to access service account tokens (P4SA) within a customer tenant project. By leveraging this identity, an attacker could theoretically read analytics metadata or tamper with internal monitoring records across other Apigee organizations (tenants).

What should I do?

Take the following actions for each affected product:

Apigee

No action is required for customers using the Google Cloud version of Apigee. Vulnerability fixes have been applied to Apigee release [1-16-0-apigee-3](#).

Apigee hybrid

To resolve this vulnerability, customers must [enable the Pub/Sub-based analytics pipeline](#) and upgrade to a supported hybrid version:

| Feature configuration | Minimum remediated version | | | Standard Apigee hybrid | v1.14.0 or higher | | | With Monetization (Mint) enabled | v1.14.3, v1.15.1, or v1.16.0 | |

High | [CVE-2025-13292](#) | | |

GCP-2026-009

Published: 2026-02-13

Description

| Description | Severity | Notes | | |

Observability Analytics user interface versions prior to January 2026 can be configured to automatically execute SQL queries. A vulnerability can allow an attacker to craft a query URL which, when opened by someone with credentials, could access table contents or incur query cost.

For more information, see the [Google Cloud Observability security bulletin](#).

| High | | |

GCP-2026-008

Published: 2026-02-10

Description

| Description | Severity | Notes | | |

A set of security vulnerabilities affect Intel® TDX firmware. These vulnerabilities encompass various flaws, including race conditions (CVE-2025-30513, CVE-2025-31944), out-of-bounds reads (CVE-2025-32007, CVE-2025-27940), use of an uninitialized variable (CVE-2025-32467), and exposure of sensitive information during transient execution (CVE-2025-27572). Collectively, these issues may allow information disclosure, escalation of privilege, or denial of service. Exploitation generally requires privileged user access on the system.

What should I do?

No customer action is required. All relevant fixes have been applied to the Google server fleet. For more information, see Intel's PSIRT technical advisory [INTEL-TA-01397](#).

| High |

- [CVE-2025-30513](#)
- [CVE-2025-31944](#)
- [CVE-2025-32007](#)

- [CVE-2025-32467](#)
- [CVE-2025-27572](#)
- [CVE-2025-27940](#)

||

GCP-2026-007

****Published:** **2026-02-09

Updated: 2026-03-25

Description

| Description | Severity | Notes | |

2026-03-25 Update: Added patch versions for Ubuntu nodes with GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2025-40297

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-40297](#) | |

GCP-2026-006

Published: 2026-01-29

Updated: 2026-02-20

Description

| Description | Severity | Notes | |

2026-02-20 Update: Added patch versions for GKE.

Multiple security vulnerabilities have been identified in the OpenSSL library. The most significant finding is CVE-2025-15467, a critical vulnerability that might allow for remote code execution (RCE) or denial of service (DoS) attacks via network-based vectors.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-15467](#) | |

GCP-2026-005

Published: 2026-01-28

Description

| Description | Severity | Notes | | |

This vulnerability affects Log Analytics interface and Cloud Monitoring dashboarding interface versions prior to January 2026.

For more information, see the [Google Cloud Observability security bulletin](#).

| High | | |

GCP-2026-004

Published: 2026-01-14

Description

| Description | Severity | Notes | | |

A vulnerability in select Arm processors impacting C4A and A4X VMs was discovered in Compute Engine:

- CVE-2025-0647

For instructions and more details, see the [Compute Engine security bulletin](#).

| Medium | [CVE-2025-0647](#) | |

GCP-2026-003

****Published:** **2026-01-09

Updated: 2026-03-25

Description

| Description | Severity | Notes | | |

2026-03-25 Update: Added patch versions for Ubuntu nodes with GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- [CVE-2025-39964](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-39964](#) | |

GCP-2026-002

****Published:** **2026-01-07

Updated: 2026-03-25

Description

| Description | Severity | Notes | |

2026-03-25 Update: Added patch versions for Ubuntu nodes with GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- [CVE-2025-40215](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-40215](#) | |

GCP-2026-001

****Published:** **2026-01-07

Description

| Description | Severity | Notes | |

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2025-40214

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-40214](#) | |

GCP-2025-076

Published: 2025-12-08

Description

| Description | Severity | Notes | |

The following vulnerability was discovered in Cloud Data Fusion:

- CVE-2025-9571

For instructions and more details, see the [Cloud Data Fusion security bulletin](#).

| High | [CVE-2025-9571](#) | |

GCP-2025-075

Published: 2025-12-07

Description

| Description | Severity | Notes | |

A vulnerability in the custom integrations feature of Google Security Operations SOAR could let an authenticated user with an *IDE role* to achieve Remote Code Execution (RCE) on the server. The vulnerability was due to insufficient validation of the Python package code, allowing an attacker to upload a package with a malicious `setup.py` file. This file could then be executed during installation, compromising of the server.

What should I do?

No customer action is required. All customers have been automatically upgraded to the fixed version: 6.3.64 or higher.

| High | [CVE-2025-13428](#) | |

GCP-2025-074

****Published:** **2025-12-05

Description

| Description | Severity | Notes | | |

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2025-39965

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-39965](#) | |

GCP-2025-073

Published:2025-12-03

Description

| Description | Severity | Notes | | |

The following vulnerabilities were discovered in Envoy Proxy:

- CVE-2025-66220
- CVE-2025-64527
- CVE-2025-64763

For instructions and more details, see the [Cloud Service Mesh security bulletin](#).

| Medium |

- [CVE-2025-66220](#)
- [CVE-2025-64527](#)
- [CVE-2025-64763](#)

| |

GCP-2025-072

****Published:** **2025-12-04

Updated: 2025-12-05

Description

| Description | Severity | Notes | |

2025-12-04 Update: A Web Application Firewall rule is now available.

A remote code execution vulnerability was found in the open-source React and Next.js frameworks. While Google Cloud itself is not vulnerable, users of these frameworks running on Google Cloud may be vulnerable.

The following framework versions are affected by this vulnerability:

- React 19.0, 19.1.0, 19.1.1, and 19.2.0
- Next.js 15.x, Next.js 16.x, and Next.js 14.3.0-canary.77 and later canary releases

The following framework versions include fixes for this vulnerability:

- React 19.0.1, 19.1.2, and 19.2.1
- Next.js 15.0.5, 15.1.9, 15.2.6, 15.3.6, 15.4.8, 15.5.7, 15.6.0-canary.58, and 16.0.7

For more information, see the [React advisory](#) and [Vercel advisory](#) on these vulnerabilities.

Customers should take immediate action to protect their workloads by redeploying their workloads with updated dependencies.

How can Google help?

A Cloud Armor web application firewall (WAF) rule designed to detect and block related exploitation attempts has been made available. This new rule is intended to help protect your internet-facing applications and services that use global or regional Application Load Balancers. We recommend deploying this rule as a temporary mitigation while you update your workloads. Instructions on applying this rule are available in [this blog post](#).

For customers using App Engine Standard, Cloud Functions, Cloud Run Functions, Cloud Run, Firebase Hosting or Firebase App Hosting, a rule is already enforced to limit exploitation through requests to custom and default domains.

Customers using Artifact Analysis will be notified of these vulnerabilities on both new and existing artifacts to help them identify at-risk workloads.

This security bulletin will be updated as new information becomes available.

| Critical | [CVE-2025-55182](#) | |

GCP-2025-071

****Published:** **2025-12-02

Updated: 2026-03-25

Description

| Description | Severity | Notes | |

**2026-03-25 Update:* * Added patch versions for Ubuntu nodes with GKE.

**2025-12-11 Update:* * Added patch versions and a severity rating for GDC (VMware).

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2025-40019

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-40019](#) | |

GCP-2025-070

****Published:** **2025-12-02

Updated: 2026-03-25

Description

| Description | Severity | Notes | |

**2026-03-25 Update:* * Added patch versions for Ubuntu nodes with GKE.

**2025-12-11 Update:* * Added patch versions and a severity rating for GDC (VMware).

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2025-40018

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-40018](#) | |

GCP-2025-069

Published: 2025-11-24

Description

| Description | Severity | Notes | |

Several security issues have been discovered in runc, an open source software component used for running containers; the attacks disclosed in these vulnerabilities (CVE-2025-31133, CVE-2025-52565, and CVE-2025-52881) allow an attacker to execute a full container breakout in your Cloud Run worker pools and jobs, leading to root privilege escalation out of the container into your sandboxed instance. An actor with privileges to deploy a malicious container image can exploit these vulnerabilities to gain access to your other containers within your Cloud Run sandboxed instance.

- CVE-2025-31133
- CVE-2025-52565
- CVE-2025-52881

Worker pools, jobs, and a limited number of services that have not yet been updated are vulnerable.

An update is being prepared for rollout in early January. This bulletin will be updated once that rollout is complete.

Please report any issues related to this vulnerability to the Cloud Vulnerability Rewards Program at g.co/vrp.

| High |

- [CVE-2025-31133](#)
- [CVE-2025-52565](#)
- [CVE-2025-52881](#)

| |

GCP-2025-068

Published: 2025-11-21

Description

| Description | Severity | Notes | |

Looker fixed a vulnerability that could have allowed an attacker with viewer permissions in Looker to craft a malicious URL that, when opened by a Looker admin, would execute an attacker-supplied script. Exploitation required at least one Looker extension installed on the instance.

Both Looker-hosted and self-hosted instances were found to be vulnerable.

This issue has already been mitigated for Looker-hosted instances.

What should I do?

Looker-hosted instances:

No customer action is required.

Self-hosted Looker instances only:

If your Looker instance is self-hosted, we recommend upgrading your Looker instances as soon as possible. This vulnerability has been patched in all supported versions of Looker for self-hosted instances. The following versions have all been updated to protect from this vulnerability:

- 25.16.0 and all later versions
- 25.12.7+
- 25.6.66+
- 25.0.79+
- 24.18.201+

You can download these Looker versions at the Looker download page: <https://download.looker.com/>

| High | [CVE-2025-12739](#) | |

GCP-2025-067

Published: 2025-11-19

Description

| Description | Severity | Notes | | |

Looker fixed a vulnerability that could have allowed an attacker to take over a Looker account in a Looker instance configured with OIDC authentication, due to email address string normalization.

Both Looker-hosted and self-hosted instances were found to be vulnerable.

This issue has already been mitigated for Looker-hosted instances.

What should I do?

Looker-hosted instances:

No customer action is required.

Self-hosted Looker instances only:

If your Looker instance is self-hosted, we recommend upgrading your Looker instances as soon as possible to one of the following versions:

- 25.10.22+
- 25.8.39+

- 25.6.57+
- 25.0.69+
- 24.18.193+
- 24.12.100+

You can download these Looker versions at the Looker download page: <https://download.looker.com/>

Note: releases 25.12 and above are not affected by this security issue.

| High | [CVE-2025-12414](#) | |

GCP-2025-066

****Published:** **2025-11-10

Updated: 2025-11-27

Description

| Description | Severity | Notes | | |

2025-11-27 Update: Added patch versions for GKE and GDC (bare metal).

A set of three container escape vulnerabilities were discovered in `runc`, the default low-level container runtime for GKE. The vulnerabilities can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- CVE-2025-31133
- CVE-2025-52565
- CVE-2025-52881

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High |

- [CVE-2025-31133](#)
- [CVE-2025-52565](#)
- [CVE-2025-52881](#)

| |

GCP-2025-065

Published: 2025-11-06

Description

| Description | Severity | Notes | | |

Researchers discovered a vulnerability in AMD SEV-SNP (Secure Nested Paging) on AMD Milan machines.

An attack could allow a malicious hypervisor to manipulate RMP initialization values potentially resulting in loss of AMD SEV-SNP guest memory integrity.

Google rolled out a mitigation that prevents this issue.

What should I do?

No customer action is needed for mitigation. Mitigations have already been applied to Confidential VM instances with AMD SEV-SNP.

However, Confidential VM instances with AMD SEV-SNP now use v4 attestation format. Customers using the [go-sev-guest](#) library to parse attestation reports should update to go-sev-guest v0.14.0 or above.

For customers using their own parsers, the attestation report format is defined by the [SEV Secure Nested Paging Firmware ABI Specification](#).

What vulnerabilities are being addressed?

For more information, see the AMD advisory [AMD-SB-3020](#).

| Medium | [CVE-2025-0033](#) | |

GCP-2025-064

****Published:** **2025-10-28

Description

| Description | Severity | Notes | | |

The following vulnerabilities were discovered in Envoy Proxy:

- CVE-2025-62504
- CVE-2025-62409

For instructions and more details, see the [Cloud Service Mesh security bulletin](#).

| Medium |

- [CVE-2025-62504](#)
- [CVE-2025-62409](#)

| |

GCP-2025-063

****Published:** **2025-10-28

Updated: 2025-11-17

Description

| Description | Severity | Notes | | |

2025-11-17 Update: Added patch versions for Ubuntu GKE nodes.

2025-10-30 Update: Added patch versions and a severity rating for GDC software for VMware

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2025-39682

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-39682](#) | |

GCP-2025-062

****Published:** **2025-10-22

Updated: 2025-10-30

Description

| Description | Severity | Notes | | |

2025-10-30 Update: Added patch versions for Ubuntu nodes on GKE, and patch versions and a severity rating for GDC software for VMware

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2024-58240

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2024-58240](#) | |

GCP-2025-061

Published: 2025-10-21

Description

| Description | Severity | Notes | | |

A remote code execution vulnerability was found in open-source Valkey and Redis. As a result, all versions that [Memorystore for Redis](#), [Memorystore for Redis Cluster](#), and [Memory store for Valkey](#) supports are impacted.

By default, Google Cloud's Memorystore assets aren't exposed to the public internet, so the risk of this vulnerability is **Low** for Memorystore users who follow [Google Cloud's security best practices](#).

What should you do?

Google has started applying patches automatically, with an estimated completion date of November 6, 2025. **No action is required from you to receive this fix.**

If you want to apply these patches to your Memorystore clusters or instances earlier than November 6, 2025, then use self-service maintenance to complete the following actions:

- View the current maintenance version of your [Memorystore for Redis instances](#), [clusters in Memorystore for Redis Cluster](#), or [Memorystore for Valkey instances](#).
- Verify if the version matches the latest patched versions.
- If the version isn't the latest maintenance version, then update your instances or clusters to the latest maintenance version using self-service maintenance for [Memorystore for Redis](#), [Memorystore for Redis Cluster](#), and [Memorystore for Valkey](#).

| Critical | [CVE-2025-49844](#) | |

GCP-2025-060

Published: 2025-10-21

Description

| Description | Severity | Notes | | |

Oracle released a [security alert](#) confirming that Oracle E-Business Suite (EBS) is vulnerable to an unauthenticated exploit.

What should I do?

Oracle EBS is not utilized by Google and Google Cloud is not impacted by this vulnerability.

| Low | [CVE-2025-61882](#) | |

GCP-2025-059

Published: 2025-10-21

Updated: 2025-10-22

Description

| Description | Severity | Notes | |

2025-10-22 Update: Added link to CVE.

On September 23, 2025, we discovered a technical issue in the Vertex AI API that resulted in a limited amount of responses being misrouted between recipients for certain third-party models when using streaming requests. This issue is now resolved. Google models, e.g. Gemini, were not impacted.

Some internal proxies did not properly handle HTTP requests that have an `Expect: 100-continue` header, resulting in a desynchronization in a streaming response connection, where a response intended for one request was instead delivered as the response for a subsequent request.

What should I do?

We have implemented fixes to properly address the presence of the `Expect: 100-continue` header, and prevent recurrence of this issue. We have also added testing, monitoring, and alerting so that we can quickly detect an occurrence of this issue to prevent regression. There is no action for customers to take at this time to prevent the unintended behavior from occurring.

The fixes were rolled out for different models on separate schedules, with Anthropic models being remediated by Sep. 26, 12:45 AM PDT and all surfaces remediated by Sep. 28, 07:10 PM PDT. Affected models on Vertex AI API and the time of resolution are listed below:

-

Anthropic [Partner Model-as-a-Service models](#) (Claude)

-

The issue has been fixed as of September 26, 2025, at 12:45 AM PDT.

-

All [Open Model-as-a-Service models](#), including: DeepSeek (R1-0528 and V3.1), OpenAI (gpt-oss-120b and gpt-oss-20b), Qwen (Next Instruct 80B, Next Thinking 80B, Qwen 3 Coder, and Qwen 3 235B), Llama (Maverick, Scout, 3.3, 3.2, 3.1 405b, 3.1 70b, and 3.1 8b)

The issue has been fixed as of September 28, 2025, at 2:43 AM PDT.

Mistral and AI21 [Partner Model-as-a-Service models](#)

The issue has been fixed as of September 28, 2025, at 11:00 AM PDT.

[Self-deployed models](#) for which the 'StreamRawPredict', 'ChatCompletions', 'GenerateContent', or 'StreamGenerateContent' method was invoked using [public endpoints](#)

The issue has been fixed as of September 28, 2025, at 7:10 PM PDT.

Neither [dedicated](#) (the default on Model Garden) nor private endpoints were impacted.

| Medium | [CVE-2025-11915](#) | |

GCP-2025-058

Published: 2025-10-20

Description

| Description | Severity | Notes | |

[A flaw has been discovered in the RDSEED instruction](#) in AMD Zen 5 processors (Turin). This instruction is used to generate cryptographic random numbers. Under certain system load conditions, the 16- and 32-bit versions of RDSEED can silently fail, which could compromise applications relying on random number generation. Customers using the 64-bit version of RDSEED are unaffected.

What should I do?

AMD is investigating the vulnerability.

It's important to note that the 64-bit Linux kernel uses the safe 64-bit version of the RDSEED instruction, and that feeds the random numbers obtained from `/dev/[u]random`. Those random numbers are not impacted by this vulnerability.

If you have application code that synthesizes random numbers itself using the RDSEED instruction, be aware that the 16-bit and 32-bit versions of the instruction are insecure.

The 64-bit version of the instruction is safe.

What vulnerabilities are being addressed?

This vulnerability allows an attacker to cause RDSEED to silently fail, potentially compromising random number generation in applications.

| High | [CVE-2025-62626](#) | |

GCP-2025-057

Published: 2025-10-17

Updated: 2025-11-11

Description

| Description | Severity | Notes | | |

2025-11-11 Update: Added patch versions for Ubuntu node pools in GKE.

2025-10-27 Update: Added patch versions and a severity rating for GDC software for VMware.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2025-38618

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-38618](#) | |

GCP-2025-056

Published: 2025-10-16

Updated: 2026-03-24

Description

| Description | Severity | Notes | | |

2026-03-24 Update: Added patch versions for Ubuntu nodes with GKE.

2025-11-13 Update: Added patch versions for GDC software for VMware. Updated severity for GDC software for VMware to High.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2025-39946

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-39946](#) | |

GCP-2025-055

****Published:** **2025-10-15

Updated: 2025-11-11

Description

| Description | Severity | Notes | |

****2025-11-11 Update:** **Added patch versions for Ubuntu node pools in GKE.

2025-10-16 Update: Added patch versions and a severity rating for GDC software for VMware

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2025-38617

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-38617](#) | |

GCP-2025-054

Published: 2025-10-14

Description

| Description | Severity | Notes | |

Per VMware security advisory [VMSA-2025-0015](#), multiple vulnerabilities in VMware Aria Operations and VMware Tools were privately reported to Broadcom. Patches are available to remediate these vulnerabilities in affected Broadcom products.

What should I do?

We recommend upgrading to VMware Aria Automation 8.18.5 and VMware Tools 13.0.5.

| Important |

- [VMSA-2025-0015](#)
- [CVE-2025-41244](#)
- [CVE-2025-41245](#)
- [CVE-2025-41246](#)

| |

GCP-2025-053

Published: 2025-09-30

Description

| Description | Severity | Notes | |

Looker Studio fixed vulnerabilities reported by an external researcher via the Google and Alphabet Vulnerability Reward Program (VRP), but found no evidence of exploitation. These issues are now resolved and no user action is required.

What should I do?

No customer action is required.

What vulnerabilities are being addressed?

The vulnerabilities allowed unauthorized access to data via shared reports, data sources with viewer credentials, Studio Linking API, and Conversational Analytics.

| High | |

GCP-2025-052

Published: 2025-09-30

Description

| Description | Severity | Notes | | |

Looker fixed vulnerabilities reported by an external researcher via the Google and Alphabet Vulnerability Reward Program (VRP), but found no evidence of exploitation. These issues are now resolved and no user action is required for Looker-hosted customers on Looker (Google Cloud core) and Looker (original). Self-hosted Looker instances are advised to update to the latest supported version.

What should I do?

Looker-hosted instances: Looker (Google Cloud core) and Looker (original) instances

No customer action is required.

Self-hosted Looker instances only

If your Looker instance is self-hosted, we recommend upgrading your Looker instances to one of the following versions:

- 25.12.30+
- 25.10.54+
- 25.6.79+
- 25.0.89+
- 24.18.209+

Note: releases 25.14 and above are not affected by these security issues.

What vulnerabilities are being addressed?

The vulnerabilities allowed users with developer permissions in Looker to access both the underlying system hosting Looker, and its internal database.

| High | |

GCP-2025-051

Published: 2025-09-11

Description

| Description | Severity | Notes | | |

Researchers discovered a security vulnerability affecting [Intel CPUs](#) and all [AMD Zen CPUs](#). This vulnerability allows an attacker to potentially read sensitive data by leveraging known speculative execution vulnerabilities.

What should I do?

No action is required at this time as Google has applied fixes to the affected assets, including Google Cloud, to mitigate the issue non-disruptively across the fleet.

What vulnerabilities are being addressed?

For more information, see [COMSEC blog post](#).

| High | [CVE-2025-40300](#) | |

GCP-2025-050

****Published:** **2025-09-10

****Updated:** **2025-11-11

Description

| Description | Severity | Notes | | |

****2025-11-11 Update:** **Added patch versions for Ubuntu node pools in GKE.

****2025-10-15 Update:** **Added patch versions and a severity rating for GDC software for VMware.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2025-38500

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-38500](#) | |

GCP-2025-049

Published: 2025-09-10

Description

| Description | Severity | Notes | | |

A critical vulnerability was found in Google Security Operations SOAR versions 6.3.54.0 and 6.3.53.2. An authenticated user with permissions to upload ZIP files (for example, when importing Use Cases) could upload a ZIP archive capable of writing files to arbitrary locations on the server's file system.

The system for extracting files from ZIP archives couldn't prevent files within the archive from being written outside their intended destination folder. This is also known as a Directory Traversal or Zip Slip vulnerability.

What should I do?

No customer action is required. All customers have been automatically upgraded to the fixed version or higher: 6.3.54.1 or 6.3.53.3

What vulnerabilities are being addressed?

An attacker could exploit this vulnerability to overwrite application files. By overwriting a JavaScript file used by the report generation feature, an attacker could achieve Remote Code Execution (RCE) on the Google SecOps SOAR instance. The attacker could run their own code on the server.

For more information, see the [Google SecOps GCP-2025-049 security bulletin](#).

| High | [CVE-2025-9918](#) | |

GCP-2025-048

****Published:** **2025-09-02

Description

| Description | Severity | Notes | | |

The following vulnerability was discovered in Envoy Proxy:

- CVE-2025-54588

For instructions and more details, see the [Cloud Service Mesh security bulletin](#).

| High | [CVE-2025-54588](#) | |

GCP-2025-047

****Published:** **2025-08-27

Updated: 2025-09-25

Description

| Description | Severity | Notes | | |

2025-09-25 Update: Added patch versions for Ubuntu node pools on GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2025-38350

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)

- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-38350](#) | |

GCP-2025-046

****Published: ****2025-08-23

****Updated: ****2025-11-11

Description

| Description | Severity | Notes | | |

****2025-11-11 Update: ****Added patch versions for Ubuntu node pools in GKE.

****2025-10-15 Update: ****Added patch versions and a severity rating for GDC software for VMware.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2025-38477

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-38477](#) | |

GCP-2025-045

****Published: ****2025-08-21

Description

| Description | Severity | Notes | | |

An external researcher reported a vulnerability in the Dataform API through the Google and Alphabet Vulnerability Reward Program (VRP). This vulnerability could potentially allow unauthorized access to customer code repositories and data. Google quickly fixed the issue and released the fix to all regions. At this time, no evidence of exploitation has been found or reported to Google.

What should I do?

No customer action is required. Google has already applied mitigations to all impacted products and services.

What vulnerabilities are being addressed?

For more information, see [CVE-2025-9118](#).

| Critical | [CVE-2025-9118](#) | |

GCP-2025-044

**Published: **2025-08-12

Description

| Description | Severity | Notes | |

Intel has notified Google of two new security vulnerabilities.

CVE-2025-21090: This vulnerability affects the following Intel processors:

- Sapphire Rapids: C3, Z3, H3, A3, v5p VM Families
- Emerald Rapids: N4, C4, M4, A3 Ultra, A4 VM Families
- Granite Rapids: N4, C4 VM Family

CVE-2025-22840: This vulnerability affects the following Intel processor:

- Granite Rapids: N4, C4 VM Family

What should I do?

No customer action is required for either vulnerability. Google will proactively update your systems during your standard and planned maintenance windows. At this time, no evidence of exploitation has been found or reported to Google.

What vulnerabilities are being addressed?

The vulnerability, [CVE-2025-21090](#), allows an unprivileged actor utilizing the AMX CPU instruction, in conjunction with the AVX CPU instruction, to render the host machine inoperative.

The vulnerability, [CVE-2025-22840](#), allows an unprivileged actor utilizing the prefetchit CPU instruction to load memory content it would otherwise not have access to, potentially leading to remote code execution.

| Medium |

- [CVE-2025-21090](#)
- [CVE-2025-22840](#)

| |

GCP-2025-043

****Published: ****2025-08-12

Description

| Description | Severity | Notes | |

A potential Denial of Service (DoS) vulnerability exists in the pure-Python implementation of `protobuf-python`. An attacker can cause a service to crash by sending a specially crafted message.

Impact

You could be affected if you import `protobuf-python` in your projects, either directly or transitively. Be aware that the Cloud Client Libraries for Python use `protobuf-python` as a dependency. If you use these libraries, make sure that you're using a version of `protobuf-python` mentioned in the following section.

What should I do?

Ensure that you're using the latest versions of the following software packages:

- `protobuf-python` (4.25.8, 5.29.5, 6.31.1)

What vulnerabilities are addressed by this patch?

The patch mitigates the following vulnerability:

The vulnerability arises from unbounded recursion when the pure-Python backend parses certain Protocol Buffers messages. An unauthenticated attacker can send a message containing deeply nested recursive groups, messages, or a series of `SGROUP` tags. This input causes the Python interpreter to exceed its maximum recursion depth, triggering a `RecursionError` that crashes the service, resulting in a Denial of Service. Any project parsing untrusted data with the affected backend is at risk.

For more information, see the [Protobuf Security Advisory](#).

|

High

CVSS v4.0

Score: 8.2

| [CVE-2025-4565](#)

This issue only affects the `python` [Protobuf implementation backend](#).

Regardless of the value that is returned when checking the Protobuf implementation backend, an upgrade is recommended as the environment variable

`PROTOCOL_BUFFERS_PYTHON_IMPLEMENTATION` can change the Protobuf implementation backend during runtime.

||

GCP-2025-042

****Published:** **2025-08-11

Description

| Description | Severity | Notes | |

[Researchers](#) discovered a security vulnerability in specific Intel CPUs, including those based on the Skylake, Broadwell, and Haswell microarchitectures. This vulnerability allows an attacker to potentially read sensitive data directly from the CPU's L1 cache that they are not authorized to access.

This vulnerability was initially disclosed in [CVE-2018-3646](#) in 2018. Upon discovery of this vulnerability, Google immediately implemented mitigations that addressed the known risks. Communication regarding the vulnerability and the initial fixes were [published at the time](#). Since then we have been researching the residual risk and working with the upstream Linux community to remediate this risk.

Recently we worked with security researchers from academia to evaluate the state of the art of CPU security mitigations, and potential attack techniques not considered back in 2018.

Google has applied fixes to the affected assets, including Google Cloud, to mitigate the issue.

What should I do?

No customer action is required. Mitigations have already been applied to the Google server fleet.

What vulnerabilities are being addressed?

For more information, see Intel advisory [INTEL-SA-00161](#) and CVE-2018-3646.

| High | [CVE-2018-3646](#) | |

GCP-2025-041

****Published:** **2025-07-21

Updated: 2025-11-10

Description

| Description | Severity | Notes | |

2025-11-10 Update: Added patch versions for GDC software for VMware. Updated severity for GDC software for VMware to High.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Ubuntu nodes:

- [CVE-2025-37890](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-37890](#) | |

GCP-2025-040

**Published: **2025-07-15

Description

| Description | Severity | Notes | |

Per advisory VMSA-2025-0013, multiple vulnerabilities in VMware ESXi were privately reported to Broadcom.

We've either already patched these vulnerabilities or are in the process of applying the necessary patches supplied by Broadcom. There are no known workarounds for these reported vulnerabilities.

Once patched, your VMware Engine deployments should be running ESXi 7.0U3w or ESXi 8.0U3f or greater.

What should I do?

Google recommends customers to monitor their workloads on VMware Engine for any unusual activities.

| Medium to Critical |

- [VMSA-2025-0013](#)
- [CVE-2025-41236](#)
- [CVE-2025-41237](#)
- [CVE-2025-41238](#)
- [CVE-2025-41239](#)

| |

GCP-2025-039

**Published: **2025-07-15

Updated: 2025-11-10

Description

| Description | Severity | Notes | |

2025-11-10 Update: Added patch versions for GDC software for VMware. Updated severity for GDC software for VMware to High.

2025-08-28 Update: Added patch versions for Ubuntu node pools on GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2025-38083

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-38083](#) | |

GCP-2025-038

****Published:** **2025-07-09

Updated: 2025-09-25

Description

| Description | Severity | Notes | |

2025-09-25 Update: Added patch versions for Ubuntu node pools on GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2025-37752

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)

- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-37752](#) | |

GCP-2025-037

****Published:** **2025-07-08

| Description | Severity | Notes | |

AMD has disclosed two vulnerabilities affecting AMD EPYC 2nd generation (Rome), 3rd generation (Milan), and 4th generation (Genoa) CPUs. The vulnerabilities allow an attacker to infer data from previous stores or the L1D cache, potentially resulting in leakage of sensitive information.

Google has rolled out a mitigation that prevents this leakage of information between VMs.

There is still the potential that processes within a single VM can exploit these vulnerabilities.

What should I do?

After July 8, 2025, a guest-level mitigation that protects against intra-guest attacks is available to the following VMs:

- VMs that are started for the first time.
- VMs that are fully stopped and started again. Operating system reboots will not enable the mitigation, and the VM itself must be fully restarted to enable the mitigation. Guest operating system kernels must support this mitigation for it to be effective.

For more information see AMD security bulletin [AMD-SB-7029](#).

| Medium |

- [CVE-2024-36350](#)
- [CVE-2024-36357](#)

| |

GCP-2025-036

****Published:** **2025-07-01

Updated: 2025-11-10

Description

| Description | Severity | Notes | |

2025-11-10 Update: Added patch versions for GDC software for VMware. Updated severity for GDC software for VMware to High.

2025-07-21 Update: Added patch versions for Ubuntu node pools on GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- [CVE-2025-38001](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-38001](#) | |

GCP-2025-035

****Published:** **2025-06-17

Updated: 2025-11-10

Description

| Description | Severity | Notes | |

2025-11-10 Update: Added patch versions for GDC software for VMware. Updated severity for GDC software for VMware to High.

2025-07-21 Update: Added patch versions for Ubuntu node pools on GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- [CVE-2025-37997](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-37997](#) | |

GCP-2025-034

****Published:** **2025-06-17

Updated: 2025-07-21

Description

| Description | Severity | Notes | | |

2025-07-21 Update: Added patch versions for Ubuntu node pools on GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2025-38000

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-38000](#) | |

GCP-2025-033

Published: 2025-06-06

Description

| Description | Severity | Notes | | |

A security issue was discovered where attackers might be able to bypass workload isolation restrictions on GKE clusters.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| Medium | N/A | |

GCP-2025-032

****Published:** **2025-06-03

Updated: 2025-07-21

Description

| Description | Severity | Notes | | |

2025-07-21 Update: Added patch versions for Ubuntu node pools on GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2025-37798

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-37798](#) | |

GCP-2025-031

****Published:** **2025-06-03

Updated: 2025-08-26

Description

| Description | Severity | Notes | | |

2025-08-26 Update: Added patch versions for Ubuntu node pools on GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2025-37797

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-37797](#) | |

GCP-2025-030

****Published: **2025-05-23**

| Description | Severity | Notes | |

Per VMware security advisory [VMSA-2024-0017](#), an SQL-injection vulnerability in VMware Aria Automation was privately reported to VMware. Patches are available to remediate this vulnerability in affected VMware products.

What should I do?

We recommend upgrading to VMware Aria Automation KB325790.

| Important |

- [VMSA-2024-0017](#)
- [CVE-2024-22280](#)

| |

GCP-2025-029

****Published: **2025-05-23**

| Description | Severity | Notes | |

Per VMware security advisory [VMSA-2025-0006](#), a local privilege escalation vulnerability in VMware Aria Operations was responsibly reported to VMware. Patches are available to remediate this vulnerability in affected VMware products.

What should I do?

We recommend upgrading to VMware Aria Operations 8.18 HF5.

| Important |

- [VMSA-2025-0006](#)
- [CVE-2025-22231](#)

| |

GCP-2025-028

****Published: **2025-05-23**

| Description | Severity | Notes | |

Per VMware security advisory [VMSA-2025-0003](#), multiple vulnerabilities in VMware Aria Operations for logs and VMware Aria Operations were privately reported to VMware. Patches are available to remediate this vulnerability in affected VMware products.

What should I do?

We recommend upgrading to VMware Aria Operations for Logs 8.18.3 and VMware Aria Operations to 8.18.3.

| Important |

- [VMSA-2025-0003](#)
- [CVE-2025-22218](#)
- [CVE-2025-22219](#)
- [CVE-2025-22220](#)
- [CVE-2025-22221](#)
- [CVE-2025-22222](#)

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GCP-2025-027

****Published: ****2025-05-16

Description

| Description | Severity | Notes | |

A security vulnerability was detected in the classic Application Load Balancer service prior to April 26, 2025.

What should I do?

No customer action is required. The issue was resolved in the Classic Application Load Balancer service on April 26, 2025.

What vulnerabilities are being addressed?

[CVE-2025-4600](#) allowed attackers to smuggle requests to classic Application Load Balancers due to incorrect parsing of oversized chunk bodies. When parsing the request body of an HTTP request using chunked transfer-encoding, the classic Application Load Balancer allows oversized chunk bodies. Consequently, it was feasible to hide bytes within this ignored trailing data that an upstream HTTP server might incorrectly interpret as a line terminator. This vulnerability was addressed within the classic Application Load Balancer service on April 26, 2025 through improved input validation and parsing logic.

We're here to help

If you have any questions or require assistance, contact [Cloud Customer Care](#).

| High | [CVE-2025-4600](#) | |

GCP-2025-026

****Published: ****2025-05-15

| Description | Severity | Notes | |

Per VMware security advisory [VMSA-2025-0008](#), a DOM based Cross-Site Scripting (XSS) vulnerability in VMware Aria Automation was privately reported to VMware. Patches are

available to remediate this vulnerability in affected VMware products.

What should I do?

We recommend upgrading to VMware Aria Automation 8.18.1 patch 2.

| Important |

- [VMSA-2025-0008](#)
- [CVE-2025-22249](#)

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GCP-2025-025

****Published: ****2025-05-13

Description

| Description | Severity | Notes | | |

Intel has notified Google about a new side channel vulnerability affecting the following Intel processors: CascadeLake, Ice Lake XeonSP, Ice Lake XeonD, Sapphire Rapids and Emerald Rapids.

Google has applied fixes to the affected assets, including Google Cloud, to ensure customers are protected. At this time, no evidence of exploitation has been found or reported to Google.

What should I do?

No customer action is required. Fixes have already been applied to the Google server fleet to protect customers.

What vulnerabilities are being addressed?

[CVE-2024-45332](#). For more information, see [Intel advisory INTEL-SA-01247](#).

We're here to help

If you have any questions or require assistance, please contact [Cloud Customer Care](#) and reference issue number 417536835.

| High | [CVE-2024-45332](#) | |

GCP-2025-024

****Published: ****2025-05-12

****Updated: ****2025-05-13

Description

| Description | Severity | Notes | | |

2025-05-13 Update: If you have any questions or require assistance, please contact [Cloud Customer Care](#) and reference issue number 417458390.

Intel has notified Google about a new speculative execution vulnerability affecting Intel Cascade Lake processors and Intel Ice Lake processors.

Google has applied fixes to the affected assets, including Google Cloud, to ensure customers are protected. At this time, no evidence of exploitation has been found or reported to Google.

What should I do?

No customer action is required. Mitigations have already been applied to the Google server fleet.

Further mitigations from Intel Original Equipment Manufacturers (OEMs) and other operating system partners will be deployed as soon as they become available to mitigate the same-mode Indirect Target Selection (ITS) vulnerability.

After the operating system mitigations have been applied, customers with long-running 3rd generation or later VMs may experience some unintended performance degradation

What vulnerabilities are being addressed?

CVE-2024-28956. For more information, see [Intel security advisory INTEL-SA-01153](#).

| High | [CVE-2024-28956](#) | |

GCP-2025-023

Published: 2025-05-05

Description

| Description | Severity | Notes | | |

Potential security gaps which could be exploited if not addressed in the JavaCallout and PythonScript policies were discovered and addressed.

For instructions and more details see the [Apigee security bulletin](#).

| High | [CVE-2023-44487](#) | |

GCP-2025-022

****Published:** **2025-05-01

Updated: 2025-05-22

Description

| Description | Severity | Notes | | |

2025-05-22 Update: Added patch versions for Ubuntu node pools on GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2025-21702

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-21702](#) | |

GCP-2025-021

****Published:** **2025-04-29

Updated: 2025-06-02

Description

| Description | Severity | Notes | | |

2025-06-02 Update: Added patch versions for Ubuntu node pools on GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2025-21971

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-21971](#) | |

GCP-2025-020

Published: 2025-04-29

Description

| Description | Severity | Notes | |

A vulnerability in Looker allowed users with admin permissions (specifically: `manage_project_connections_restricted`) in Looker to read files from the underlying host filesystem and query internal network endpoints. This issue is now resolved, and no user action is required for Looker-hosted customers using Looker (Google Cloud core) and Looker (original). Self-hosted Looker instances are advised to update to the latest supported version.

This vulnerability has been patched in all supported versions of customer-hosted Looker, which are available on the [Looker download page](#).

What should I do?

- For all Looker-hosted instances, including both Looker (Google Cloud core) and Looker (original) instances, there are no actions you need to take.
- For Looker customer-hosted instances, please update to the latest supported version of Looker as soon as possible. The versions below have all been updated to protect from this vulnerability. You can download these versions at the Looker download page:
- 25.4 -> 25.4.29+
- 25.2 -> 25.2.34+
- 25.0 -> 25.0.55+
- 24.18 -> 24.18.185+
- 24.12 -> 24.12.95+
- 24.6 -> 24.6.107+

| High | | |

GCP-2025-019

****Published:** **2025-04-25

Updated: 2025-06-26

Description

| Description | Severity | Notes | |

2025-06-26 Update: Added patch versions for GDC software for VMware.

2025-05-22 Update: Added patch versions for Ubuntu node pools on GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- [CVE-2025-21701](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-21701](#) | |

GCP-2025-018

****Published:** **2025-04-23

Updated: 2025-10-09

Description

| Description | Severity | Notes | | |

2025-10-09 Update: added patch versions for Ubuntu nodes on GKE

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- [CVE-2025-40364](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-40364](#) | |

GCP-2025-017

****Published:** **2025-04-17

Updated: 2025-05-22

Description

| Description | Severity | Notes | | |

2025-05-22 Update: Added patch versions for Ubuntu node pools on GKE.

2025-05-05 Update: Added patch versions for GDC software for VMware. Updated severity for GDC software for VMware to High.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- [CVE-2025-21756](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-21756](#) | |

GCP-2025-016

****Published:** **2025-04-16

Updated: 2025-05-22

Description

| Description | Severity | Notes | |

2025-05-22 Update: Added patch versions for Ubuntu node pools on GKE.

2025-04-29 Update: Added patch version for GDC software for VMware. Updated severity for GDC software for VMware to High.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- [CVE-2023-52927](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2023-52927](#) | |

GCP-2025-015

****Published: ****2025-04-15

Updated: 2025-05-22

Description

| Description | Severity | Notes | |

2025-05-22 Update: Added patch versions for Ubuntu node pools on GKE.

2025-04-17 Update: Added patch versions for GDC (VMware). Updated GDC (VMware) severity from Pending to High.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2025-21700

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-21700](#) | |

GCP-2025-014

****Published: ****2025-04-10

Updated: 2025-05-22

Description

| Description | Severity | Notes | |

2025-05-22 Update: Added patch versions for Ubuntu node pools on GKE.

2025-05-05 Update: Added patch versions for GDC software for VMware. Updated severity for GDC software for VMware to High.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2025-21703

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)

- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2025-21703](#) | |

GCP-2025-013

Published: 2025-03-24

Description

| Description | Severity | Notes | | |

Several security issues have been discovered in the NGINX Ingress Controller, `ingress-nginx`, an open source software component that runs inside Kubernetes clusters to help manage network traffic coming into the cluster. The most critical is CVE-2025-1974. For the full details and a complete list, see the [Kubernetes CVE feed](#).

These issues affect `ingress-nginx`. If you do not have `ingress-nginx` installed on your cluster, you are not affected.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| None | [CVE-2025-1974](#) | |

GCP-2025-012

****Published:** **2025-03-19

Updated: 2025-04-10

Description

| Description | Severity | Notes | | |

2025-04-10 Update: Added patch versions for Ubuntu GKE nodes and GDC software for VMware. Updated severity for GDC software for VMware to High.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2024-53164

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2024-53164](#) | |

GCP-2025-011

****Published:** **2025-03-06

| Description | Severity | Notes | | |

VMware disclosed multiple vulnerabilities in [VMSA-2025-0004](#) that impact ESXi components deployed in customer environments.

VMware Engine impact

Your private clouds are either already patched or are in the process of being updated to address the security vulnerability. As part of the VMware Engine service, all customers get dedicated bare metal hosts with local attached disks that are physically isolated from other hardware. This means that the vulnerability is scoped to guest VMs within your specific private cloud only.

Your private clouds will be updated to 7.0u3s Build number 24534642. This is equivalent to 7.0U3s: Build number 24585291.

What should I do?

Follow instructions from Broadcom and your security vendors regarding this vulnerability.

| Critical |

- [VMSA-2025-0004](#)
- [CVE-2025-22224](#)
- [CVE-2025-22225](#)
- [CVE-2025-22226](#)

| |

GCP-2025-010

****Published:** **2025-03-05

Updated: 2025-06-02

Description

| Description | Severity | Notes | | |

2025-06-02 Update: Added patch versions for Ubuntu node pools on GKE.

2025-04-10 Update: Added patch versions for GDC software for VMware. Updated severity for GDC software for VMware to High.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2024-56770

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2024-56770](#) | |

GCP-2025-009

Published: 2025-03-05

Description

| Description | Severity | Notes | | |

The Envoy project recently announced several new security vulnerabilities ([CVE-2024-53269](#), [CVE-2024-53270](#), and [CVE-2024-53271](#)) that could allow an attacker to crash Envoy.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High |

- [CVE-2024-53269](#)
- [CVE-2024-53270](#)
- [CVE-2024-53271](#)

| |

GCP-2025-008

****Published:** **2025-02-19

Updated: 2025-04-10

Description

| Description | Severity | Notes | |

2025-04-10 Update: Added patch versions for GDC software for VMware. Updated severity for GDC software for VMware to High.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- CVE-2024-53141

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2024-53141](#) | |

GCP-2025-007

Published: 2025-02-03

Description

| Description | Severity | Notes | |

Google has discovered a vulnerability in AMD Zen-based CPUs that affects Confidential VM instances with AMD SEV-SNP enabled. This vulnerability allows attackers with root access in a physical machine to compromise the confidentiality and integrity of the Confidential VM instance.

Google has applied fixes to the affected assets, including Google Cloud, to ensure customers are protected. At this time, no evidence of exploitation has been found or reported to Google.

What should I do?

No customer action is required. Customers who want to verify the fix can check the Trusted Computing Base (TCB) version in the attestation report from their Confidential VM instance with AMD SEV-SNP. The minimum versions that mitigate this vulnerability are as follows:

```
SNP TCB SVN: 0x18 0d24
tcb_version {
  psp_bootloader_version: 4
  snp_firmware_version: 24 (0x18)
  microcode_version: 219
}
```

For more information, see AMD security bulletin [AMD-SB-3019](#).

| High |

[CVE-2024-56161](#)

||

GCP-2025-006

Published: 2025-01-23

Description

| Description | Severity | Notes | |

A vulnerability in the [Google Secret Manager Provider for Secret Store CSI Driver](#) allows an attacker with Pod and Secret creation permissions in a namespace to exfiltrate the CSI driver's Kubernetes service account token by mounting a malicious volume on a Pod.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| Medium | |

GCP-2025-005

Published: 2025-01-22

Description

| Description | Severity | Notes | |

Servers that use Google authentication libraries and Cloud Client Libraries to authenticate to Google Cloud with a credential configuration controlled by an attacker could be susceptible to server-side request forgery and arbitrary file reads.

What should I do?

If you accept a credential configuration (credential JSON, file, or stream) from an external source for authentication to Google Cloud, you must validate it before providing it to Google authentication libraries or Cloud Client Libraries. Providing an unvalidated credential configuration to Google libraries can compromise the security of your systems and data. For more information, see [Validate credential configurations from external source S](#).

What vulnerabilities are being addressed?

Some types of credential configurations include endpoints and file paths, which the authentication libraries use to acquire a token. When a service or application accepts externally sourced credential configurations and uses them with Google authentication libraries or Cloud Client Libraries without validation, an attacker can provide a credential configuration that contains a malicious endpoint or path. This allows the attacker to exfiltrate data or tokens from the service or the machine the service is running on. To prevent this, validations should be performed on the externally sourced credential configuration as described in [Validate credential configurations from external sources](#).

To inform application developers, we have updated our documentation with the validations to perform when accepting credential configurations that are obtained from external sources.

| High | | |

GCP-2025-004

****Published:** **2025-01-16

| Description | Severity | Notes | | |

Per VMware security advisory VMSA-2025-0001, a server-side request forgery (SSRF) vulnerability in VMware Aria Automation was responsibly reported to VMware. Patches are available to remediate this vulnerability in affected VMware products.

What should I do?

We recommend upgrading to VMware Aria Automation 8.18.2 HF.

| Medium |

- [VMSA-2025-0001](#)
- [CVE-2025-22215](#)

| |

GCP-2025-003

****Published:** **2025-01-09

Updated: 2025-01-23

Description

| Description | Severity | Notes | | |

2025-01-23 Update: Added patch versions for Ubuntu node pools on GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2024-50264

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2024-50264](#) | |

GCP-2025-002

****Published:** **2025-01-09

Updated: 2025-01-23

Description

| Description | Severity | Notes | | |

2025-01-23 Update: Added patch versions for Ubuntu node pools on GKE.

2025-01-22 Update: Added patch versions for GDC (VMware). Updated severity for GDC (VMware) to Medium.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2024-53057

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2024-53057](#) | |

GCP-2025-001

Published: 2025-01-08

Updated: 2025-01-23

Description

| Description | Severity | Notes | | |

2025-01-23 Update: Updated the **Affected resources** section in the [GKE](#) tab.

2025-01-08 Update: Corrected the start date and time of the issue.

A security issue impacted resources in VPCs with GKE Multi-Cluster Gateway (MCG) configured. MCG is an optional feature that is used by a small subset of GKE customers. We are individually notifying customers who had the feature enabled during that time period.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | None | |

GCP-2024-065

Description

| Description | Severity | Notes | | |

The following CVEs expose Cloud Service Mesh to exploitable vulnerabilities:

- CVE-2024-53269: Happy Eyeballs: Validate that additional_address are IP addresses instead of crashing when sorting.
- CVE-2024-53270: HTTP/1: Sending overload crashes when the request is reset beforehand.
- CVE-2024-53271: HTTP/1.1 Multiple issues with envoy.reloadable_features.http1_balsa_delay_reset.

For instructions and more details, see the [Cloud Service Mesh security bulletin](#).

| High |

- [CVE-2024-53269](#)
- [CVE-2024-53270](#)

- [CVE-2024-53271](#)

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GCP-2024-064

****Published: ****2024-12-10

| Description | Severity | Notes | | |

Per VMware security advisory VMSA-2024-0022, multiple vulnerabilities in VMware Aria Operations were responsibly reported to VMware. Updates are available to remediate these vulnerabilities in the affected VMware product.

What should I do?

We recommend upgrading to VMware Aria Operations 8.18.2.

| Important |

- [VMSA-2024-0022](#)
- [CVE-2024-38830](#)
- [CVE-2024-38831](#)
- [CVE-2024-38832](#)
- [CVE-2024-38833](#)
- [CVE-2024-38834](#)

||

GCP-2024-063

****Published: ****2024-12-06

| Description | Severity | Notes | | |

A vulnerability was discovered in the Vertex AI API serving Gemini multimodal requests, allowing bypass of [VPC Service Controls](#). An attacker may be able to abuse the `fileUri` parameter of the API to exfiltrate data.

What should I do?

No actions needed. We've implemented a fix to return an error message when a media file URL is specified in the `fileUri` parameter and VPC Service Controls is enabled. Other use cases are unaffected.

What vulnerabilities are being addressed?

The Cloud Support API serving Gemini multimodal requests lets you include media files by specifying the URL of the media file in the `fileUri` parameter. This capability can be used to bypass VPC Service Controls perimeters. An attacker inside the service perimeter could encode sensitive data in the `fileUri` parameter to bypass the service perimeter.

| Medium | [CVE-2024-12236](#) | |

GCP-2024-062

****Published:** **2024-12-02

Updated: 2025-01-22

Description

| Description | Severity | Notes | |

2025-01-22 Update: Added patch versions for GDC (VMware). Updated GDC (VMware) severity from Pending to High.

****2024-12-12 Update:** **Added patch versions for Ubuntu node pools on GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- CVE-2024-46800

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2024-46800](#) | |

GCP-2024-061

Published: 2024-11-25

Description

| Description | Severity | Notes | |

A security issue discovered in Kubernetes clusters could result in remote code execution using a `gitRepo` volume. If the git repository is maliciously constructed, a user with the ability to create a Pod and associate a `gitRepo` volume could execute arbitrary commands beyond the container boundary.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2024-10220](#) | |

GCP-2024-060

****Published: ****2024-10-17

| Description | Severity | Notes | | |

Per VMware security advisory VMSA-2024-0020, multiple vulnerabilities in VMware NSX were responsibly reported to VMware.

The version NSX-T running on your VMware Engine environment is not impacted by CVE-2024-38815, CVE-2024-38818, or CVE-2024-38817.

What should I do?

Because VMware Engine clusters are not affected by this vulnerability, no further action is required.

| Medium |

- [VMSA-2024-0020](#)
- [CVE-2024-38815](#)
- [CVE-2024-38817](#)
- [CVE-2024-38818](#)

| |

GCP-2024-059

****Published: ****2024-10-16

| Description | Severity | Notes | | |

Per VMware security advisory VMSA-2024-0021, an authenticated SQL injection vulnerability in VMware HCX was privately reported to VMware.

We have applied the mitigation approved by VMware to address this vulnerability. This fix addresses a security vulnerability described in CVE-2024-38814. The image versions running in your VMware Engine private cloud don't reflect any change at this time to indicate the changes applied. Appropriate mitigations have been installed and your environment is secured from this vulnerability.

What should I do?

We recommend upgrading to VMware HCX version 4.9.2.

| High |

- [VMSA-2024-0021](#)
- [CVE-2024-38814](#)

| |

GCP-2024-058

Published: 2024-10-16

Description

| Description | Severity | Notes | | |

Migrate to Containers for Windows versions 1.1.0 to 1.2.2 created a local `m2cuser` with administrator privileges. This posed a security risk if the `analyze` or `generate` commands were interrupted by the user or due to an internal error causing skipping the action to delete the local user `m2cuser`.

What should I do?

The following versions of Migrate to Containers CLI for Windows have been updated with code to fix this vulnerability. We recommend that you manually upgrade your Migrate to Containers CLI to the following version or higher:

- Migrate to Containers CLI for Windows 1.2.3 released on October 8, 2024 - [Migrate to Containers CLI release notes](#) | [Google Cloud](#)

What vulnerabilities are being addressed?

The vulnerability, CVE-2024-9858, allows an attacker to gain administrator access to impacted Windows machines using the local administrator user created by the Migrate to Containers software.

| Medium | [CVE-2024-9858](#) | |

GCP-2024-057

****Published:** **2024-10-03

Updated: 2024-11-19

Description

| Description | Severity | Notes | | |

****2024-11-19 Update:** **Added patch versions for Ubuntu node pools on GKE.

2024-10-15 Update: Added patch versions for GDC (VMware). Updated GKE and GDC (VMware) severity levels.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- CVE-2024-45016

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)

- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| Medium | [CVE-2024-45016](#) | |

GCP-2024-056

Published: 2024-09-27

Description

| Description | Severity | Notes | | |

A vulnerability chain (CVE-2024-47076, CVE-2024-47175, CVE-2024-47176, CVE-2024-47177) that could result in remote code execution was discovered in the CUPS printing system used by some Linux distributions. An attacker can exploit this vulnerability if the CUPS services are listening on UDP port 631 and they can connect to it.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| None |

- [CVE-2024-47076](#)
- [CVE-2024-47175](#)
- [CVE-2024-47176](#)
- [CVE-2024-47177](#)

| |

GCP-2024-055

Published: 2024-09-24

Description

| Description | Severity | Notes | | |

An HTTP Request Smuggling vulnerability in Looker allowed an unauthorized attacker to capture HTTP responses destined for legitimate users.

There are two Looker versions that are hosted by Looker:

- Looker (Google Cloud core) was found to be vulnerable. This issue has already been mitigated and our investigation has found no signs of exploitation.
- Looker (original) was not vulnerable to this issue.

Customer-hosted Looker instances were found to be vulnerable and must be upgraded to one of the versions below.

This vulnerability has been patched in all supported versions of customer-hosted Looker, which are available on the [Looker download page](#).

What should I do?

- For all Looker-hosted instances, including Looker (Google Cloud core) instances, there are no actions you need to take.
- For Looker customer-hosted instances, please update to the latest supported version of Looker as soon as possible. The versions below have all been updated to protect from this vulnerability. You can download these versions at the Looker download page:
- 23.12 -> 23.12.123+
- 23.18 -> 23.18.117+
- 24.0 -> 24.0.92+
- 24.6 -> 24.6.77+
- 24.8 -> 24.8.66+
- 24.10 -> 24.10.78+
- 24.12 -> 24.12.56+
- 24.14 -> 24.14.37+

What vulnerabilities are being addressed?

The vulnerability, CVE-2024-8912, allows an attacker to send crafted HTTP request headers to Looker, possibly resulting in the interception of HTTP responses destined for other users.

These responses could contain sensitive information.

This vulnerability is only exploitable under certain specific configurations.

| Medium | [CVE-2024-8912](#) | |

GCP-2024-054

Published: 2024-09-23

Description

| Description | Severity | Notes | |

A security issue was discovered in Kubernetes clusters with Windows nodes where `BUILTIN\Users` may be able to read container logs and `NT AUTHORITY\Authenticated Users` may be able to modify container logs.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| Medium | [CVE-2024-5321](#) | |

GCP-2024-053

Published: 2024-09-19

Description

| Description | Severity | Notes | |

When parsing unknown fields in the Protobuf Java Full and Lite libraries, a maliciously crafted message can cause a StackOverflow error and lead to a program crash.

What should I do?

We have been working diligently to address this issue and have released a mitigation that is available now. We encourage that you're using the latest versions of the following software packages:

- protobuf-java (3.25.5, 4.27.5, 4.28.2)
- protobuf-javalite (3.25.5, 4.27.5, 4.28.2)
- protobuf-kotlin (3.25.5, 4.27.5, 4.28.2)
- protobuf-kotlin-lite (3.25.5, 4.27.5, 4.28.2)
- com-protobuf [JRuby gem only] (3.25.5, 4.27.5, 4.28.2)

What vulnerabilities are addressed by this patch?

This vulnerability is a potential Denial of Service.

Parsing nested groups as unknown fields with DiscardUnknownFieldsParser or Java Protobuf Lite parser, or against Protobuf map fields, creates unbounded recursions that can be abused by an attacker.

|

CVSS4.0 score 8.7

High

| [CVE-2024-7254](#) | |

GCP-2024-052

Description

| Description | Severity | Notes | |

The following CVEs expose Cloud Service Mesh to exploitable vulnerabilities:

- CVE-2024-45807: oghttp2 crash on OnBeginHeadersForStream
- CVE-2024-45808: Malicious log injection via access logs
- CVE-2024-45806: Potential to manipulate `x-envoy` headers from external sources
- CVE-2024-45809: JWT filter crash in the clear route cache with remote JWKS
- CVE-2024-45810: Envoy crashes for LocalReply in http async client

For instructions and more details, see the [Cloud Service Mesh security bulletin](#).

| Medium to High |

- [CVE-2024-45807](#)
- [CVE-2024-45808](#)
- [CVE-2024-45806](#)
- [CVE-2024-45809](#)
- [CVE-2024-45810](#)

| |

GCP-2024-051

****Published: ****2024-09-18

| Description | Severity | Notes | |

VMware disclosed multiple vulnerabilities in [VMSA-2024-0019](#) that impact vCenter components deployed in customer environments.

VMware Engine impact

- Google has already disabled any potential exploit of this vulnerability. For example, Google has blocked the ports through which this vulnerability could be exploited.
- In addition, Google ensures all future deployments of vCenter are not exposed to this vulnerability.

What should I do?

No further action is required at this time.

| Critical |

- [CVE-2024-38812](#)
- [CVE-2024-38813](#)

| |

GCP-2024-050

Published: 2024-09-04

Description

| Description | Severity | Notes | | |

A new remote code execution vulnerability (CVE-2024-38063) has been discovered in Windows. An attacker could remotely exploit this vulnerability by sending specially crafted IPv6 packets to a host.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| None | [CVE-2024-38063](#) | |

GCP-2024-049

****Published:** **2024-08-21

Updated: 2024-11-01

Description

| Description | Severity | Notes | | |

****2024-11-01 Update:** **Added patch versions for Ubuntu node pools on GKE.

2024-10-21 Update: Added patch versions and updated Severity for GDC (VMware).

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- CVE-2024-36978

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2024-36978](#) | |

GCP-2024-048

****Published:** **2024-08-20

Updated: 2024-10-30

Description

| Description | Severity | Notes | | |

****2024-10-30 Update:** **Added patch versions for Ubuntu node pools on GKE.

2024-10-25 Update: Added patch versions and updated Severity for GDC (VMware).

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- CVE-2024-41009

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2024-41009](#) | |

GCP-2024-047

****Published:** **2024-08-19

Updated: 2024-10-30

Description

| Description | Severity | Notes | | |

****2024-10-30 Update:** **Added patch versions for Ubuntu node pools on GKE.

2024-10-21 Update: Added patch versions and updated Severity for GDC (VMware).

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- CVE-2024-39503

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)

- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2024-39503](#) | |

GCP-2024-046

****Published:** **2024-08-05

Description

| Description | Severity | Notes | | |

AMD has notified Google about 3 new (2 medium risk, 1 high risk) firmware vulnerabilities affecting SEV-SNP in AMD EPYC 3rd generation (Milan) and 4th generation (Genoa) CPUs.

Google has applied fixes to the affected assets, including Google Cloud, to ensure customers are protected. At this time, no evidence of exploitation has been found or reported to Google.

What should I do?

No customer action is required. Fixes have already been applied to the Google server fleet. For more information, see AMD security bulletin [AMD-SN-3011](#).

| Medium-High |

[CVE-2023-31355](#)

[CVE-2024-21978](#)

[CVE-2024-21980](#)

| |

GCP-2024-045

****Published:** **2024-07-17

Updated: 2024-09-19

Description

| Description | Severity | Notes | | |

**2024-09-19 Update:* * Added patch versions for GDC software for VMware.

2024-08-21 Update: Added patch versions for Ubuntu node pools on GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- [CVE-2024-26925](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2024-26925](#) | |

GCP-2024-044

****Published:** **2024-07-16

Updated: 2024-10-30

Description

| Description | Severity | Notes | |

****2024-10-30 Update:** **Added patch versions for Ubuntu node pools on GKE.

2024-10-21 Update: Added patch versions and updated Severity for GDC (VMware).

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- [CVE-2024-36972](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2024-36972](#) | |

GCP-2024-043

****Published:** **2024-07-16

Updated: 2024-10-02

Description

| Description | Severity | Notes | | |

**2024-10-02 Update:* * Added patch versions for Ubuntu node pools on GKE.

**2024-09-20 Update:* * Added patch versions for GDC software for VMware.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- CVE-2024-26921

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2024-26921](#) | |

GCP-2024-042

****Published:** **2024-07-15

Updated: 2024-07-18

Description

| Description | Severity | Notes | | |

2024-07-18 Update: Clarified that Autopilot clusters in the default configuration aren't impacted.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- CVE-2024-26809

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2024-26809](#) | |

GCP-2024-041

****Published:** **2024-07-08

Updated: 2024-09-16

Description

| Description | Severity | Notes | | |

**2024-09-16 Update:* * Added patch versions for GDC software for VMware.

**2024-07-19 Update:* * Added patch versions for Ubuntu node pools on GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- CVE-2023-52654
- CVE-2023-52656

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High |

- [CVE-2023-52654](#)
- [CVE-2023-52656](#)

| |

GCP-2024-040

Published: 2024-07-01

Updated: 2024-07-16

Description

| Description | Severity | Notes | | |

2024-07-16 Updates:

Some Serverless VPC Access customers are potentially impacted by a vulnerability in OpenSSH (CVE-2024-6387). In the case of successful exploit, this could allow a remote, unauthenticated attacker to execute arbitrary code as root on the target virtual machine.

Exploitation is believed to be difficult. For example, customers can't access the VMs and the VMs don't have public IPs. We are not aware of any exploitation attempts.

What should I do?

Serverless VPC Access deployments have been automatically updated by Google where possible. However, you should verify that the [Google-managed service agent has its required role](#). If not, your Serverless VPC Access connector might still be vulnerable. We recommend that you [migrate to Direct VPC egress](#), or deploy a new connector and delete the old connector, to ensure that you have the required update with the fix.

2024-07-11 Update: Added patch versions for GDC software for VMware, GKE on AWS, and GKE on Azure. For details, in the GKE documentation, see the following bulletins:

- [GDC software for VMware security bulletin](#)
 - [GKE on AWS security bulletin](#)
 - [GKE on Azure security bulletin](#)
-

2024-07-10 Update:

- Added security bulletin for Migrate to Virtual Machines.
-

2024-07-09 Updates:

Some App Engine flexible environment customers are potentially impacted by a vulnerability in OpenSSH (CVE-2024-6387). In the case of successful exploit, this could allow a remote, unauthenticated attacker to execute arbitrary code as root on the target virtual machine.

What should I do?

Google has already updated flexible environment deployments automatically where possible. However some customers who [disabled the Google-managed service agent](#), or made changes to the Google Cloud APIs or other default configurations, could not be updated and might still be vulnerable. You should deploy a new version of your app to pick up the update with the fix.

Please note that updated deployments will report ssh version `OpenSSH_9.6p1`. This version has been patched with a fix for CVE-2024-6387.

What vulnerabilities are being addressed?

The vulnerability CVE-2024-6387, which allows a remote, unauthenticated attacker to execute arbitrary code as root on the target machine.

2024-07-08 Updates:

Managed Service for Apache Spark clusters on Google Compute Engine running on image version 2.2 (all operating systems) and 2.1 (Debian only) are impacted by a vulnerability in

OpenSSH (CVE-2024-6387) which in case of successful exploit could allow a remote, unauthenticated attacker to execute arbitrary code as root on the target machine.

Managed Service for Apache Spark on Google Compute Engine image versions 2.0 and 1.5, as well as Managed Service for Apache Spark images version 2.1 not running on Debian, are not impacted. Managed Service for Apache Spark clusters with Personal Authentication enabled are not impacted. Managed Service for Apache Spark Serverless is also not impacted.

What should I do?

Please update your Managed Service for Apache Spark clusters on Google Compute Engine to one of the following versions:

- 2.2.24 or later
- 2.1.58 or later

In case you are not able to update your Managed Service for Apache Spark clusters to one of the versions above, we recommend using the initialization action available at this location: `gs://dataproc-initialization-actions/hotfixes/openssh-CVE-2024-6387-mitigation.sh`

Please follow these instructions on how to [specify initialization actions for Managed Service for Apache Spark](#). Please note that the initialization action has to be run on every node (masters and workers) for pre-existing clusters.

2024-07-03 Updates:

- Added patch versions for GKE.
- Added security bulletin for GDC connected.

2024-07-02 Updates:

- Clarified that Autopilot clusters are impacted and will require user action.
- Added impact assessments and mitigation steps for GDC software for VMware, GKE on AWS, and GKE on Azure.
- Corrected the GDC software for bare metal security bulletin to clarify that GDC software for bare metal isn't directly affected and that customers should check with OS vendors for patches.

A remote code execution vulnerability, [CVE-2024-6387](#), was recently discovered in OpenSSH. The vulnerability exploits a race condition that can be used to obtain access to a remote shell, enabling attackers to gain root access. At the time of publication, exploitation is believed to be difficult and take several hours per machine being attacked. We are not aware of any exploitation attempts.

For instructions and more details, see the following bulletins:

- [Compute Engine security bulletin](#)

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)
- [Google Cloud VMware Engine security bulletin](#)
- [Apigee security bulletin](#)
- [Dataflow security bulletin](#)
- [GDC connected security bulletin](#)
- [Migrate to Virtual Machines security bulletin](#)

| Critical | [CVE-2024-6387](#) | |

GCP-2024-039

****Published:** **2024-06-28

Updated: 2024-09-25

Description

| Description | Severity | Notes | | |

**2024-09-25 Update:* * Added patch versions for GDC software for VMware.

2024-08-20 Update: Added patch versions for Ubuntu node pools on GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- [CVE-2024-26923](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2024-26923](#) | |

GCP-2024-038

****Published:** **2024-06-26

Updated: 2024-09-17

Description

| Description | Severity | Notes | |

*2024-09-17 Update: * Added patch versions for GDC software for VMware.

2024-08-06 Update: Added patch versions for Ubuntu node pools on GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- CVE-2024-26924

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GDC software for VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GDC software for bare metal security bulletin](#)

| High | [CVE-2024-26924](#) | |

GCP-2024-037

**Published: **2024-06-18

| Description | Severity | Notes | |

VMware disclosed multiple vulnerabilities in [VMSA-2024-0012](#) that impact vCenter components deployed in customer environments.

Google Cloud VMware Engine impact

- The vulnerability can be exploited by accessing specific ports in vCenter Server. Google has already blocked the vulnerable ports on vCenter server, which prevents any potential exploits of this vulnerability.
- In addition, Google ensures all future deployments of vCenter are not exposed to this vulnerability.

What should I do?

No further action is required at this time.

| Critical |

- [CVE-2024-37079](#)
- [CVE-2024-37080](#)
- [CVE-2024-37081](#)

| |

GCP-2024-036

****Published:** **2024-06-18

Description

| Description | Severity | Notes | | |

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2024-26584

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2024-26584](#) | |

GCP-2024-035

****Published:** **2024-06-12

Updated: 2024-07-18

Description

| Description | Severity | Notes | | |

****2024-07-18 Update:** **Added patch versions for Ubuntu node pools on GKE and added a patch version for version 1.27 on Container-Optimized OS node pools.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- CVE-2024-26584

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2024-26584](#) | |

GCP-2024-034

****Published:** **2024-06-11

Updated: 2024-07-10

Description

| Description | Severity | Notes | | |

2024-07-10 Update: Added patch versions for Container-Optimized OS nodes running minor version 1.26 and 1.27 and added patch versions for Ubuntu nodes.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2024-26583

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2024-26583](#) | |

GCP-2024-033

****Published:** **2024-06-10

Updated: 2024-09-26

Description

| Description | Severity | Notes | | |

**2024-09-26 Update:* * Added patch versions for GDC software for VMware.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes:

- CVE-2022-23222

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)

- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2022-23222](#) | |

GCP-2024-032

Published: 2024-06-04

Description

| Description | Severity | Notes | | |

The following CVEs expose Cloud Service Mesh to exploitable vulnerabilities:

- CVE-2024-23326: Envoy incorrectly accepts HTTP 200 response for entering upgrade mode.
- CVE-2024-32974: Crash in EnvoyQuicServerStream::OnInitialHeadersComplete().
- CVE-2024-32975: Crash in QuicheDataReader::PeekVarInt62Length().
- CVE-2024-32976: Endless loop while decompressing Brotli data with extra input.
- CVE-2024-34362: Crash (use-after-free) in EnvoyQuicServerStream.
- CVE-2024-34363: Crash due to uncaught nlohmann JSON exception.
- CVE-2024-34364: Envoy OOM vector from HTTP async client with unbounded response buffer for mirror response.

For instructions and more details, see the [Cloud Service Mesh security bulletin](#).

| High |

- [CVE-2024-23326](#)
- [CVE-2024-32974](#)
- [CVE-2024-32975](#)
- [CVE-2024-32976](#)
- [CVE-2024-34362](#)
- [CVE-2024-34363](#)
- [CVE-2024-34364](#)

| |

GCP-2024-031

Published: 2024-05-24

Description

| Description | Severity | Notes | | |

A new vulnerability (CVE-2024-4323) has been discovered in Fluent Bit that could result in remote code execution. Fluent Bit versions 2.0.7 through 3.0.3 are affected.

GKE, GKE on VMware, GKE on AWS, GKE on Azure, and GKE on Bare Metal don't use a vulnerable version of Fluent Bit and are **unaffected**.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| None | [CVE-2024-4323](#) | |

GCP-2024-030

****Published:** **2024-05-15

Updated: 2024-07-18

Description

| Description | Severity | Notes | |

****2024-07-18 Update:** **Added patch versions for Ubuntu node pools on GKE

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- CVE-2023-52620

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2023-52620](#) | |

GCP-2024-029

****Published:** **2024-05-14

Updated: 2024-08-19

Description

| Description | Severity | Notes | | |

2024-08-19 Update: Added patch versions for Ubuntu node pools on GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- CVE-2024-26642

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2024-26642](#) | |

GCP-2024-028

****Published:** **2024-05-13

Updated: 2024-05-22

Description

| Description | Severity | Notes | | |

2024-05-22 Update: Added patch versions for Ubuntu

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- CVE-2024-26581

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2024-26581](#) | |

GCP-2024-027

****Published:** **2024-05-08

Updated: 2024-09-25

Description

| Description | Severity | Notes | |

**2024-09-25 Update:* * Added patch versions for GDC software for VMware.

2024-05-15 Update: Added patch versions for GKE Ubuntu node pools.

2024-05-09 Update: Corrected severity from Medium to High and clarified that GKE Autopilot clusters in the default configuration are not impacted.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- CVE-2024-26808

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2024-26808](#) | |

GCP-2024-026

****Published:** **2024-05-07

Updated: 2024-08-06

Description

| Description | Severity | Notes | |

2024-08-06 Update: Added patch versions for Ubuntu node pools on GKE.

2024-05-09 Update: Corrected severity from Medium to High.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- CVE-2024-26643

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2024-26643](#) | |

GCP-2024-025

Published: 2024-04-26

Description

| Description | Severity | Notes | | |

Looker fixed vulnerabilities reported by an external researcher via the Google and Alphabet Vulnerability Reward Program (VRP) program, but found no evidence of exploitation. These issues are now resolved and no user action is required for Looker-hosted customers on Looker (Google Cloud core) and Looker (original). Self-hosted Looker instances are advised to update to the latest supported version.

What should I do?

Looker-hosted instances: Looker (Google Cloud core) and Looker (original) instances

No customer action is required.

Self-hosted Looker instances only

If your Looker instance is self-hosted, we recommend upgrading your Looker instances to one of the following [versions](#):

- 24.6.12+
- 24.4.27+
- 24.2.58+
- 24.0.65+
- 23.18.100+
- 23.12.105+
- 23.6.163+

How was this fixed?

Google disabled direct administrative access to the internal database from the Looker application, removed elevated privileges that enabled cross-tenant access, and rotated the exposed secrets. Additionally, we have patched path traversal vulnerabilities that potentially exposed service account credentials. We are also conducting a thorough review of our code and systems to identify and address any similar potential vulnerabilities.

| Critical | | |

GCP-2024-024

****Published:** **2024-04-25

Updated: 2024-07-18

Description

| Description | Severity | Notes | | |

****2024-07-18 Update:** **Added patch versions for Ubuntu node pools on GKE

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- CVE-2024-26585

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2024-26585](#) | |

GCP-2024-023

Published: 2024-04-24

Description

| Description | Severity | Notes | | |

The following CVEs expose Cloud Service Mesh to exploitable vulnerabilities:

- CVE-2024-27919: HTTP/2: memory exhaustion due to CONTINUATION frame flood.
- CVE-2024-30255: HTTP/2: CPU exhaustion due to CONTINUATION frame flood
- CVE-2024-32475: Abnormal termination when using 'auto_sni' with ':authority' header longer than 255 characters.
- CVE-2023-45288: HTTP/2 CONTINUATION frames can be utilized for DoS attacks.

For instructions and more details, see the [Cloud Service Mesh security bulletin](#).

| High |

- [CVE-2024-27919](#)
- [CVE-2024-30255](#)
- [CVE-2024-32475](#)

- [CVE-2023-45288](#)

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GCP-2024-022

Published: 2024-04-03

Updated: 2024-07-17

Description

| Description | Severity | Notes | |

2024-07-17 Update: Added patch versions for GKE on VMware

2024-07-09 Update: Added patch versions for GKE on Bare Metal

2024-04-24 Update: Added patch versions for GKE.

A Denial-of-Service (DoS) vulnerability (CVE-2023-45288) was recently discovered in multiple implementations of the HTTP/2 protocol, including the golang HTTP server used by Kubernetes. The vulnerability could lead to a DoS of the Google Kubernetes Engine (GKE) control plane.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2023-45288](#) | |

GCP-2024-021

****Published:** **2024-04-03

Description

| Description | Severity | Notes | |

Compute Engine is not affected by [CVE-2024-3094](#), which affects versions 5.6.0 and 5.6.1 of the xz-utils package in the liblzma library, and could lead to compromise of the OpenSSH utility.

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For more details, see the [Compute Engine security bulletin](#).

Medium | [CVE-2024-3094](#) |

GCP-2024-020

****Published:** **2024-04-02

Description

| Description | Severity | Notes | |

Researchers discovered a vulnerability ([CVE-2023-48022](#)) in [Ray](#). Ray is a [third-party, open source tool](#) for AI workloads. Because Ray does not require authentication, threat actors can achieve remote code execution through submitting jobs to publicly exposed instances. The vulnerability has been [disputed by Anyscale](#), the developer of Ray. Ray maintains its functions are an intended, core product feature, and that security must instead be implemented outside of a Ray cluster, as any unintended network exposure of the Ray cluster could lead to compromise.

Based on the response, this CVE is disputed and may not show up in vulnerability scanners. Regardless, it is being actively exploited in the wild and users should configure their usage as suggested below.

What should I do?

Follow Ray [best practices](#) and guidelines, including running trusted code on trusted networks, in order to secure your Ray workloads. Deployment of [ray.io](#) in customer cloud instances falls under the model of shared responsibility.

Google Kubernetes Engine (GKE) security has published a [blog](#) on hardening Ray on GKE.

For further information on ways to add authentication and authorization to Ray services, consult the [Identity-Aware Proxy \(IAP\) documentation](#). GKE users can implement IAP following this [guidance](#) or by repurposing Terraform modules linked in the [blog](#).

| High | [CVE-2023-48022](#) | |

GCP-2024-018

****Published:** **2024-03-12

Updated: 2024-04-04, 2024-05-06

Description

| Description | Severity | Notes | |

2024-05-06 Update: Added patch versions for GKE Ubuntu node pools.

2024-04-04 Update: Corrected minimum versions for GKE Container-Optimized OS node pools.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- CVE-2024-1085

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2024-1085](#) | |

GCP-2024-017

****Published: ****2024-03-06

Description

| Description | Severity | Notes | | |

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- CVE-2023-3611

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2023-3611](#) | |

GCP-2024-016

****Published: ****2024-03-05

| Description | Severity | Notes | | |

VMware disclosed multiple vulnerabilities in [VMSA-2024-0006](#) that impact ESXi components deployed in customer environments.

Google Cloud VMware Engine impact

Your private clouds have been updated to address the security vulnerability.

What should I do?

No action is needed on your part.

| Critical |

- [VMSA-2024-0006](#)
- [CVE-2024-22252](#)
- [CVE-2024-22253](#)
- [CVE-2024-22254](#)
- [CVE-2024-22255](#)

| |

GCP-2024-014

**Published: **2024-02-26

Description

| Description | Severity | Notes | | |

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- [CVE-2023-3776](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2023-3776](#) | |

GCP-2024-013

**Published: **2024-02-27

Description

| Description | Severity | Notes | | |

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- [CVE-2023-3610](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2023-3610](#) | |

GCP-2024-012

****Published: ****2024-02-20

Description

| Description | Severity | Notes | | |

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- [CVE-2024-0193](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2024-0193](#) | |

GCP-2024-011

****Published: ****2024-02-15

Description

| Description | Severity | Notes | | |

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes:

- [CVE-2023-6932](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)

- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2023-6932](#) | |

GCP-2024-010

****Published:** **2024-02-14

Updated: 2024-04-17

Description

| Description | Severity | Notes | | |

2024-04-17 Update: Added patch versions for GKE on VMware.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes.

- CVE-2023-6931

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2023-6931](#) | |

GCP-2024-009

Published: 2024-02-13

Description

| Description | Severity | Notes | | |

On February 13, 2024, AMD disclosed two vulnerabilities affecting SEV-SNP on EPYC CPUs based on third generation "Milan" and fourth generation "Genoa" Zen cores. The vulnerabilities allow privileged attackers to access stale data from guests or cause a loss of guest integrity.

Google has applied fixes to affected assets, including Google Cloud, to ensure customers are protected. At this time, no evidence of exploitation has been found or reported to Google.

What should I do?

No customer action is required. Fixes have already been applied to the Google server fleet for Google Cloud, including Compute Engine.

For more information, see AMD security bulletin [AMD-SN-3007](#).

| Moderate |

- [CVE-2023-31346](#)
- [CVE-2023-31347](#)

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GCP-2024-008

Published: 2024-02-12

Description

| Description | Severity | Notes | | |

CVE-2023-5528 allows an attacker to create pods and persistent volumes on Windows nodes in a way that enables admin privilege escalation on those nodes.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2023-5528](#) | |

GCP-2024-007

Published: 2024-02-08

Description

| Description | Severity | Notes | | |

The following CVEs expose Cloud Service Mesh to exploitable vulnerabilities:

- CVE-2024-23322: Envoy crashes when idle and requests per try timeout occur within the backoff interval.
- CVE-2024-23323: Excessive CPU usage when URI template matcher is configured using regex.
- CVE-2024-23324: External authorization can be bypassed when Proxy protocol filter sets invalid UTF-8 metadata.

- Envoy crashes when using an address type that isn't supported by the OS.
- CVE-2024-23327: Crash in proxy protocol when command type is `LOCAL`.

For instructions and more details, see the [Cloud Service Meshsecurity bulletin](#).

| High |

- [CVE-2024-23322](#)
- [CVE-2024-23323](#)
- [CVE-2024-23324](#)
- [CVE-2024-23325](#)
- [CVE-2024-23327](#)

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GCP-2024-006

Published: 2024-02-5

Description

| Description | Severity | Notes | | |

When an Apigee API Management proxy connects to a [target endpoint](#) or [target server](#), the proxy does not perform hostname validation for the certificate presented by the target endpoint or target server by default. If hostname validation is not enabled using one of the following options, Apigee proxies connecting to a target endpoint or target server may be at risk for a man-in-the-middle attack by an authorized user. For more information, see [Configuring TLS from Edge to the backend \(Cloud and Private Cloud\)](#).

Apigee proxy deployments on the following Apigee platforms are affected:

- Apigee Edge for Public Cloud
- Apigee Edge for Private Cloud

For instructions and more details, see the [Apigee security bulletin](#).

| High | | |

GCP-2024-005

Published: 2024-01-31 **Updated:** 2024-04-02, 2024-05-06

Description

| Description | Severity | Notes | | |

2024-05-06 Update: Added patch versions for GKE on AWS and GKE on Azure.

2024-04-02 Update: Added patch versions for GKE on Bare Metal

2024-03-06 Update: Added patch versions for GKE on VMware

2024-02-28 Update: Added patch versions for Ubuntu

2024-02-15 Update: Clarified that the 1.25 and 1.26 Ubuntu patch versions in the 2024-02-14 update might cause unhealthy nodes.

2024-02-14 Update: Added patch versions for Ubuntu

**2024-02-06 Update:* * Added patch versions for Container-Optimized OS.

A security vulnerability, CVE-2024-21626, has been discovered in `runc` where a user with permission to create Pods on Container-Optimized OS and Ubuntu nodes might be able to gain full access to the node filesystem.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2024-21626](#) | |

GCP-2024-004

Published: **2024-01-24 ****Updated:** 2024-02-07

Description

| Description | Severity | Notes | |

**2024-02-07 Update:* * Added patch versions for Ubuntu.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes.

- CVE-2023-6817

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)

- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2023-6817](#) | |

GCP-2024-003

Published: 2024-01-19 **Updated:** 2024-01-26

Description

| Description | Severity | Notes | | |

2024-01-26 Update: Clarified the number of affected clusters and the actions that we took to help mitigate the impact. For details, see the [GCP-2024-003 security bulletin](#).

We have identified several clusters where users have granted Kubernetes privileges to the `system:authenticated` group, which includes all users with a Google account. These types of bindings are [not recommended](#), as they violate the principle of least privilege and grant access to very large groups of users. See guidance under '[What should I do](#)' for instructions on how to find these types of bindings.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)

| Medium | | |

GCP-2024-002

****Published:** **2024-01-17

Updated: 2024-02-20

Description

| Description | Severity | Notes | | |

2024-02-20 Update: Added patch versions for GKE on VMware.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS nodes.

- CVE-2023-6111

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)

- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2023-6111](#) | |

GCP-2024-001

Published: 2024-01-09

Description

| Description | Severity | Notes | | |

Several vulnerabilities were discovered in the TianoCore EDK II UEFI firmware. This firmware is used in Google Compute Engine VMs. If exploited, the vulnerabilities could allow a bypass of secure boot, which would provide false measurements in the secure boot process, including when used in Shielded VMs.

What should I do?

No action is required. Google has patched this vulnerability across Compute Engine and all VMs are protected from this vulnerability.

What vulnerabilities are addressed by this patch?

The patch mitigated the following vulnerabilities:

- CVE-2022-36763
- CVE-2022-36764
- CVE-2022-36765

| Medium |

- [CVE-2022-36763](#)
- [CVE-2022-36764](#)
- [CVE-2022-36765](#)

| |

GCP-2023-051

****Published:** **2023-12-28

Description

| Description | Severity | Notes | | |

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes.

- [CVE-2023-3609](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2023-3609](#) | |

GCP-2023-050

****Published: ****2023-12-27

Description

| Description | Severity | Notes | | |

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes.

- [CVE-2023-3389](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2023-3389](#) | |

GCP-2023-049

****Published: ****2023-12-20

Description

| Description | Severity | Notes | | |

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes.

- [CVE-2023-3090](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)

- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2023-3090](#) | |

GCP-2023-048

****Published:** **2023-12-15

Updated: 2023-12-21

Description

| Description | Severity | Notes | | |

2023-12-21 Update: Clarify that GKE Autopilot clusters in the default configuration are not impacted.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes.

- [CVE-2023-3390](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2023-3390](#) | |

GCP-2023-047

Published: 2023-12-14

Description

| Description | Severity | Notes | | |

An attacker who has compromised the Fluent Bit logging container could combine that access with high privileges required by Cloud Service Mesh (on clusters that have enabled it) to escalate privileges in the cluster.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)

- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| Medium | |

GCP-2023-046

Published: **2023-11-22 ****Updated:** 2024-03-04

Description

| Description | Severity | Notes | |

**2024-03-04 Update:* * Added GKE versions for GKE on VMware.

2024-01-22 Update: Added Ubuntu patch versions

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes.

- CVE-2023-5717

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [GKE on Bare Metal security bulletin](#)

| High | [CVE-2023-5717](#) | |

GCP-2023-045

****Published:** **2023-11-20

Updated: 2023-12-21

Description

| Description | Severity | Notes | |

2023-12-21 Update: Clarify that GKE Autopilot clusters in the default configuration are not impacted.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes.

- CVE-2023-5197

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2023-5197](#) | |

GCP-2023-044

Published: 2023-11-15

Description

| Description | Severity | Notes | |

On November 14, AMD disclosed multiple vulnerabilities that impact various AMD server CPUs. Specifically, the vulnerabilities impact EPYC Server CPUs leveraging Zen core generation 2 "Rome," gen 3 "Milan," and gen 4 "Genoa."

Google has applied fixes to affected assets, including Google Cloud, to ensure customers are protected. At this time, no evidence of exploitation has been found or reported to Google.

What should I do?

No customer action is required.

Fixes have already been applied to the Google server fleet for Google Cloud, including Google Compute Engine.

What vulnerabilities are being addressed?

The patch mitigated the following vulnerabilities:

- CVE-2022-23820
- CVE-2021-46774
- CVE-2023-20533
- CVE-2023-20519
- CVE-2023-20592
- CVE-2023-20566
- CVE-2023-20521
- CVE-2021-46766
- CVE-2022-23830
- CVE-2023-20526
- CVE-2021-26345

For more information, see AMD's security advisory [AMD-SN-3005: "AMD INVD Instruction Security Notice"](#), also published as CacheWarp, and [AMD-SN-3002: "AMD Server Vulnerabilities – November 2023"](#).

| Moderate |

- [CVE-2022-23820](#)
- [CVE-2021-46774](#)
- [CVE-2023-20533](#)
- [CVE-2023-20519](#)
- [CVE-2023-20592](#)
- [CVE-2023-20566](#)
- [CVE-2022-23830](#)
- [CVE-2023-20526](#)
- [CVE-2021-26345](#)

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GCP-2023-043

Published: 2023-11-14

Description

| Description | Severity | Notes | |

Intel disclosed a CPU vulnerability in select processors. Google has taken steps to mitigate its server fleet, including Google Compute Engine for Google Cloud, and Chrome OS devices to ensure customers are protected.

The vulnerability details:

- [CVE-2023-23583](#)

What should I do?

No customer action is required.

The mitigation provided by Intel for the affected processors has been applied to the Google server fleet, including Google Compute Engine for Google Cloud.

At this time, Google Distributed Cloud Edge requires an update from the OEM. Google will remediate this product once the update has been made available, and this bulletin will be updated accordingly.

Chrome OS devices with the affected processors received the fix automatically as part of releases 119, 118, and 114 (LTS).

What vulnerabilities are being addressed?

[CVE-2023-23583](#). For details, see [Intel Security Advisory INTEL-SA-00950](#).

| High | [CVE-2023-23583](#) | |

GCP-2023-042

Published: **2023-11-13 ****Updated:** 2023-11-15

Description

| Description | Severity | Notes | |

2023-11-15 Update: Clarify that only the listed minor versions need to upgrade to a corresponding patched version for GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes.

- CVE-2023-4147

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2023-4147](#) | |

GCP-2023-041

****Published:** **2023-11-08

Updated: 2023-11-21, 2023-12-05, 2023-12-21

Description

| Description | Severity | Notes | |

2023-12-21 Update: Clarify that GKE Autopilot clusters in the default configuration are not impacted.

2023-12-05 Update: Added additional GKE versions for Container-Optimized OS node pools.

2023-11-21 Update: Clarify that only the listed minor versions need to upgrade to a corresponding patched version for GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes.

- [CVE-2023-4004](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2023-4004](#) | |

GCP-2023-040

****Published:** **2023-11-06

Updated: 2023-11-21, 2023-12-21

Description

| Description | Severity | Notes | | |

2023-12-21 Update: Clarify that GKE Autopilot clusters in the default configuration are not impacted.

2023-11-21 Update: Clarify that only the listed minor versions need to upgrade to a corresponding patched version for GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes.

- [CVE-2023-4921](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2023-4921](#) | |

GCP-2023-039

Published: 2023-11-06

****Updated:** **2023-11-21, 2023-11-16

Description

| Description | Severity | Notes | |

2023-11-21 Update: Clarify that only the listed minor versions need to upgrade to a corresponding patched version for GKE.

2023-11-16 Update: The vulnerability associated with this security bulletin is [CVE-2023-4622](#). CVE-2023-4623 was incorrectly listed as the vulnerability in a previous version of the security bulletin.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes.

- CVE-2023-4623

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2023-4622](#) | |

GCP-2023-038

****Published:** **2023-11-06

Updated: 2023-11-21, 2023-12-21

Description

| Description | Severity | Notes | |

2023-12-21 Update: Clarify that GKE Autopilot clusters in the default configuration are not impacted.

2023-11-21 Update: Clarify that only the listed minor versions need to upgrade to a corresponding patched version for GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes.

- CVE-2023-4623

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2023-4623](#) | |

GCP-2023-037

Published: 2023-11-06

Updated: 2023-11-21, 2023-12-21

Description

| Description | Severity | Notes | |

2023-12-21 Update: Clarify that GKE Autopilot clusters in the default configuration are not impacted.

2023-11-21 Update: Clarify that only the listed minor versions need to upgrade to a corresponding patched version for GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes.

- [CVE-2023-4015](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2023-4015](#) | |

GCP-2023-036

Published: 2023-10-30

Description

| Description | Severity | Notes | |

[Deep Learning VM Images](#) is a set of prepackaged virtual machine images with a deep learning framework that are ready to be run out of the box. Recently, an out-of-bounds write vulnerability was discovered in the `ReadHuffmanCodes()` function in the `libwebp` library. This might impact images that use this library.

Google Cloud continuously scans its publicly published images and updates the packages to assure patched distros are included in the latest releases available for customer adoption. Deep Learning VM Images have been updated to ensure that the latest VM images include the patched distros. Customers adopting the latest VM images are not exposed to this vulnerability.

What should I do?

Google Cloud customers using published VM images should ensure that they are adopting the latest images and that their environments are up to date as per the [shared responsibility model](#).

[CVE-2023-4863](#) could be exploited by an attacker to execute arbitrary code. This vulnerability was identified in Google Chrome prior to 116.0.5845.187 and in `libwebp` prior to 1.3.2 and is being listed under CVE-2023-4863.

| High | [CVE-2023-4863](#) | |

GCP-2023-035

Published: 2023-10-26

Updated: 2023-11-21, 2023-12-21

Description

| Description | Severity | Notes | |

2023-12-21 Update: Clarify that GKE Autopilot clusters in the default configuration are not impacted.

2023-11-21 Update: Clarify that only the listed minor versions need to upgrade to a corresponding patched version for GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes.

- CVE-2023-4206
- CVE-2023-4207
- CVE-2023-4208
- CVE-2023-4128

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)

- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2023-4206](#)[CVE-2023-4207](#), [CVE-2023-4208](#), [CVE-2023-4128](#) | |

GCP-2023-034

Published: 2023-10-25

Updated: 2023-10-27

Description

| Description | Severity | Notes | | |

VMware disclosed multiple vulnerabilities in [VMSA-2023-0023](#) that impact vCenter components deployed in customer environments.

Cloud Customer Care impact

- The vulnerability can be exploited by accessing specific ports in vCenter Server. These ports are not exposed to the public internet.
- If your vCenter ports 2012/tcp, 2014/tcp, and 2020/tcp are not accessible by untrusted systems, then you are not exposed to this vulnerability.
- Google has already blocked the vulnerable ports on vCenter server, preventing any potential exploit of this vulnerability.
- In addition, Google will ensure all future deployments of vCenter server are not exposed to this vulnerability.
- At the time of the bulletin, VMware is not aware of any exploitation "in the wild". For more details please refer to the [VMware documentation](#) for more information.

What should I do?

No further action is required at this time

| Critical | [CVE-2023-34048](#),[CVE-2023-34056](#) | |

GCP-2023-033

Published: 2023-10-24

Updated: 2023-11-21, 2023-12-21

Description

| Description | Severity | Notes | | |

2023-12-21 Update: Clarify that GKE Autopilot clusters in the default configuration are not impacted and GKE Sandbox workloads are not impacted.

2023-11-21 Update: Clarify that only the listed minor versions need to upgrade to a corresponding patched version for GKE.

The following vulnerabilities were discovered in the Linux kernel that can lead to a privilege escalation on Container-Optimized OS and Ubuntu nodes.

- [CVE-2023-3777](#)

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2023-3777](#) | |

GCP-2023-032

Published: 2023-10-13

Updated: 2023-11-03

Description

| Description | Severity | Notes | |

2023-11-03 Update: Added [known issue](#) for Apigee Edge for Private Cloud.

A Denial-of-Service (DoS) vulnerability was recently discovered in multiple implementations of the HTTP/2 protocol (CVE-2023-44487), including the Apigee Ingress (Cloud Service Mesh) service used by Apigee X and Apigee Hybrid. The vulnerability could lead to a DoS of Apigee API management functionality.

For instructions and more details see the [Apigee security bulletin](#).

| High | [CVE-2023-44487](#) | |

GCP-2023-031

Published: 2023-10-10

Description

| Description | Severity | Notes | |

A denial of service attack can affect the data plane when using the HTTP/2 protocol. For instructions and more details, see the [Cloud Service Mesh security bulletin](#).

| High | [CVE-2023-44487](#) | |

GCP-2023-030

Published: 2023-10-10

Updated: 2024-03-20

Description

| Description | Severity | Notes | | |

2024-03-20 Update: Added patch versions for GKE on AWS and GKE on Azure with the latest patches for CVE-2023-44487.

2024-02-14 Update: Added patch versions for GKE on VMware.

2023-11-09 Update: Added CVE-2023-39325. Updated GKE versions with the latest patches for CVE-2023-44487 and CVE-2023-39325.

A Denial-of-Service (DoS) vulnerability was recently discovered in multiple implementations of the HTTP/2 protocol (CVE-2023-44487), including the golang HTTP server used by Kubernetes. The vulnerability could lead to a DoS of the Google Kubernetes Engine (GKE) control plane. GKE clusters with authorized networks configured are protected by limiting network access, but all other clusters are affected.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2023-44487](#), [CVE-2023-39325](#) | |

GCP-2023-029

Published: 2023-10-03

Description

| Description | Severity | Notes | | |

TorchServe is used to host PyTorch machine learning models for online prediction. Vertex AI provides prebuilt PyTorch model serving containers which depend on TorchServe. Vulnerabilities were recently discovered in TorchServe which would allow an attacker to take control of a TorchServe deployment if its model management API is exposed. Customers with PyTorch models deployed to Vertex AI online prediction are not affected by these vulnerabilities, since Vertex AI does not expose TorchServe's model management API. Customers using TorchServe outside of Vertex AI should take precautions to ensure their deployments are set up securely.

What should I do?

Vertex AI customers with deployed models using Vertex AI's [prebuilt PyTorch serving containers](#) do not need to take any action to address the vulnerabilities, since Vertex AI's deployments do not expose TorchServe's management server to the internet.

Customers who are using the prebuilt PyTorch containers in other contexts, or who are using a custom-built or third-party distribution of TorchServe, should do the following:

- Ensure that TorchServe's model management API is not exposed to the internet. The model management API can be restricted to local access only by ensuring that the `[management_address]` (<https://pytorch.org/serve/configuration.html#configure-torchserve-listening-address-and-port>) is bound to `127.0.0.1`.
- Use the `[allowed_urls]` (<https://pytorch.org/serve/configuration.html#other-properties>) setting to ensure that models can be loaded from intended sources only.
- Upgrade TorchServe to version 0.8.2, which includes mitigations for this issue, as soon as possible. As a precaution, Vertex AI will release fixed prebuilt containers by 2023-10-13.

What vulnerabilities are being addressed?

TorchServe's management API is bound to `0.0.0.0` by default in most TorchServe Docker images, including those released by Vertex AI, making it accessible to external requests. The default IP address for the management API is changed to `127.0.0.1` in TorchServe 0.8.2, mitigating this issue.

[CVE-2023-43654](#) and [CVE-2022-1471](#) allow a user with access to the management API to load models from arbitrary sources and remotely execute code. Mitigations for both of these issues are included in TorchServe 0.8.2: the remote code execution path is removed, and a warning is emitted if the default value for `allowed_urls` is used.

| High | [CVE-2023-43654](#), [CVE-2022-1471](#) | |

GCP-2023-028

Published: 2023-09-19

****Updated:** **2024-05-29

Description

| Description | Severity | Notes | | | **2024-05-29 Update:** The new feeds no longer use the shared service account, but it remains active for existing feeds to avoid service disruptions. Changes to the source in older feeds are blocked to prevent misuse of the shared service account. Customers can continue using their old feeds normally, as long as they don't change the source.

Customers can configure Google Security Operations to ingest data from customer-owned Cloud Storage buckets using an ingestion feed. Until recently, Google Security Operations provided a shared service account that customers used to grant permission to the bucket. An opportunity existed such that one customer's Google Security Operations instance could be configured to ingest data from another customer's Cloud Storage bucket. After performing an impact analysis, we found no current or prior exploitation of this vulnerability. The vulnerability was present in all versions of Google Security Operations prior to Sept 19, 2023.

What should I do?

As of Sept 19, 2023, Google Security Operations has been updated to address this vulnerability. No customer action is required.

What vulnerabilities are being addressed?

Previously, Google Security Operations provided a shared service account that customers used to grant permission to a bucket. Because different customers gave the same Google Security Operations service account permission to their bucket, an exploitation vector existed that allowed one customer's feed to access a different customer's bucket when a feed was being created or modified. This exploitation vector required knowledge of the bucket URI. Now, during feed creation or modification, Google Security Operations uses unique service accounts for each customer.

| High | | |

GCP-2023-027

Published: 2023-09-11

| Description | Severity | Notes | | |

VMware vCenter Server updates address multiple memory corruption vulnerabilities (CVE-2023-20892, CVE-2023-20893, CVE-2023-20894, CVE-2023-20895, CVE-2023-20896)

Customer Care impact

VMware vCenter Server (vCenter Server) and VMware Cloud Foundation (Cloud Foundation).

What should I do?

Customers are not impacted and no action needs to be taken.

| Medium |

- [CVE-2023-20893](#)

||

GCP-2023-026

Published: 2023-09-06

Description

| Description | Severity | Notes | |

Three vulnerabilities (CVE-2023-3676, CVE-2023-3955, CVE-2023-3893) have been discovered in Kubernetes where a user that can create Pods on Windows nodes may be able to escalate to admin privileges on those nodes. These vulnerabilities affect the Windows versions of Kubelet and the Kubernetes CSI proxy.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2023-3676](#), [CVE-2023-3955](#), [CVE-2023-3893](#) | |

GCP-2023-025

Published: 2023-08-08

| Description | Severity | Notes | |

Intel recently announced [Intel Security Advisory INTEL-SA-00828](#) impacting some of their processor families. You are encouraged to assess your risks based on the advisory.

Google Cloud VMware Engine impact

Our fleet utilizes the impacted processor families. In our deployment, the entire server is dedicated to one customer. Hence, our deployment model doesn't add any additional risk to your assessment of this vulnerability.

We are working with our partners to obtain necessary patches and will be deploying these patches on priority across the fleet using the standard upgrade process in the next several weeks.

What should I do?

No action is needed on your part, we are working on upgrading all the impacted systems.

| High |

- [CVE-2022-40982](#)

GCP-2023-024

Published: 2023-08-08

Updated: 2023-08-10, 2024-06-04

Description

| Description | Severity | Notes | |

2024-06-04 Update: The following missing products have now been updated to fix this vulnerability:

- Google Distributed Cloud Hosted
- Google Distributed Cloud Edge

2023-08-10 Update: Added ChromeOS LTS version number.

Intel disclosed a vulnerability in select processors (CVE-2022-40982). Google has taken steps to mitigate its server fleet, including Google Cloud, to ensure customers are protected.

The vulnerability details:

- CVE-2022-40982 (Intel IPU 2023.3, "GDS" aka "Downfall")

What should I do?

No customer action is required.

All available patches have already been applied to the Google server fleet for Google Cloud, including Google Compute Engine.

At this time, the following products require additional updates from partners and vendors.

- Google Cloud VMware Engine
- Google Distributed Cloud Hosted
- Google Distributed Cloud Edge
- Google Cloud Bare Metal Solution
- Evolved Packet Core

Google will remediate these products once these patches have been made available, and this bulletin will be updated accordingly.

Google Chromebook and ChromeOS Flex customers automatically received the Intel provided mitigations in Stable (115), LTS (108), Beta (116), and LTC (114). Chromebook and ChromeOS Flex customers pinned to an older release should consider unpinning and moving to Stable or LTS releases to ensure they receive this and other vulnerability fixes.

What vulnerabilities are being addressed?

CVE-2022-40982 - For more information, see [Intel Security Advisory INTEL-SA-00828](#).

| High | [CVE-2022-40982](#) | |

GCP-2023-023

Published: 2023-08-08

Description

| Description | Severity | Notes | |

AMD disclosed a vulnerability in select processors (CVE-2023-20569). Google has taken steps to mitigate its server fleet, including Google Cloud, to ensure customers are protected.

The vulnerability details:

- CVE-2023-20569 (AMD SB-7005 aka "Inception")

What should I do?

Users of Compute Engine VMs should consider OS provided mitigations if using intra-instance untrusted code execution. We recommend customers to contact their OS vendors for more specific guidance.

Fixes have already been applied to the Google server fleet for Google Cloud, including Google Compute Engine.

What vulnerabilities are being addressed?

CVE-2023-20569 - For more information, see [AMD SB-7005](#).

| Moderate | [CVE-2023-20569](#) | |

GCP-2023-022

****Published:** **2023-08-03

Description

| Description | Severity | Notes | |

Google identified a vulnerability in gRPC C++ Implementations prior to the 1.57 release. This was a Denial-of-Service vulnerability within the gRPC's C++ implementation. These have been fixed in the 1.53.2, 1.54.3, 1.55.2, 1.56.2, and 1.57 releases.

What should I do?

Ensure that you're using the latest versions of the following software packages:

- gRPC (C++, Python, Ruby) versions 1.53, 1.54, 1.55, and 1.56 need to upgrade to the following patch releases:

- 1.53.2
- 1.54.3
- 1.55.2
- 1.56.2
- gRPC (C++, Python, Ruby) versions 1.52 and earlier need to upgrade to one of the approved patch releases. For example, 1.53.2, 1.54.3, 1.53.4, etc.

What vulnerabilities are being addressed?

These patches mitigate the following vulnerabilities:

- Denial-Of-Service vulnerability in gRPC C++ implementations: Specially crafted requests can cause a termination of connection between a proxy and a backend.

High | [CVE-2023-33953](#) | |

GCP-2023-021

Updated:2023-07-26

Published:2023-07-25

Description

| Description | Severity | Notes | |

The following CVEs expose Cloud Service Mesh to exploitable vulnerabilities:

- CVE-2023-35941: A malicious client is able to construct credentials with permanent validity in some specific scenarios. For example, the combination of host and expiration time in the HMAC payload can be always valid in OAuth2 filter's HMAC check.
- CVE-2023-35942: gRPC access loggers using the listener's global scope can cause a use-after-free crash when the listener is drained. This can be triggered by an LDS update with the same gRPC access log configuration.
- CVE-2023-35943: If `origin` header is configured to be removed with `request_headers_to_remove: origin`, CORS filter will `segfault` and crash Envoy.
- CVE-2023-35944: Attackers can send mixed scheme requests to bypass scheme checks in Envoy. For example, if a request with mixed scheme HTTP is sent to the OAuth2 filter, it will fail the exact-match checks for HTTP, and inform the remote endpoint the scheme is HTTPS, thus potentially bypassing OAuth2 checks specific to HTTP requests.

For instructions and more details, see the [Cloud Service Mesh security bulletin](#).

| High |

- [CVE-2023-35941](#)
- [CVE-2023-35942](#)
- [CVE-2023-35943](#)
- [CVE-2023-35944](#)

GCP-2023-020

Updated:2023-07-26

****Published:** **2023-07-24

Description

| Description | Severity | Notes | | |

AMD has released a [microcode update](#) that addresses a hardware security vulnerability (CVE-2023-20593). Google has applied the necessary fixes for this vulnerability to its server fleet, including servers for the Google Cloud Platform. Testing indicates there is no impact to the performance of systems.

What should I do?

No customer action is required, as fixes have already been applied to the Google server fleet for Google Cloud Platform.

What vulnerabilities are being addressed?

CVE-2023-20593 addresses a vulnerability in some AMD CPUs. More information can be found [here](#).

High | [CVE-2023-20593](#) | | |

GCP-2023-019

Published:2023-07-18

Description

| Description | Severity | Notes | | |

A new vulnerability (CVE-2023-35945) has been discovered in Envoy where a specifically crafted response from an untrusted upstream service can cause a denial of service through memory exhaustion. This is caused by Envoy's HTTP/2 codec which may leak a header map and bookkeeping structures upon receiving `RST_STREAM` immediately followed by the `GOAWAY` frames from an upstream server.

For instructions and more details, see the [Cloud Service Mesh security bulletin](#).

High | [CVE-2023-35945](#) | | |

GCP-2023-018

Published: 2023-06-27

Description

| Description | Severity | Notes | |

A new vulnerability (CVE-2023-2235) has been discovered in the Linux kernel that can lead to a privilege escalation on the node. GKE Autopilot clusters are affected as GKE Autopilot nodes always use Container-Optimized OS node images. GKE Standard clusters with versions 1.25 or later that are running Container-Optimized OS node images are affected.

GKE clusters are not affected if they are running only Ubuntu node images, or running versions before 1.25, or using [GKE Sandbox](#).

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2023-2235](#) | |

GCP-2023-017

Published: 2023-06-26

Updated: 2023-07-11

Description

| Description | Severity | Notes | |

2023-07-11 Update: New GKE versions have been updated to include the latest Ubuntu versions that patch CVE-2023-31436.

A new vulnerability (CVE-2023-31436) has been discovered in the Linux kernel that can lead to a privilege escalation on the node. GKE clusters, including Autopilot clusters, are affected. GKE clusters using [GKE Sandbox](#) are not affected.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2023-31436](#) | |

GCP-2023-016

Published: 2023-06-26

Description

| Description | Severity | Notes | | |

A number of vulnerabilities have been discovered in Envoy, which is used in Cloud Service Mesh that allow a malicious attacker to cause a denial of service or crash Envoy. These were reported separately as [GCP-2023-002](#).

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2023-27496](#), [CVE-2023-27488](#), [CVE-2023-27493](#), [CVE-2023-27492](#), [CVE-2023-27491](#), [CVE-2023-27487](#) | |

GCP-2023-015

Published: 2023-06-20

Description

| Description | Severity | Notes | | |

A new vulnerability, CVE-2023-0468, has been discovered in the Linux kernel that could allow an unprivileged user to escalate privileges to root when io_poll_get_ownership will keep increasing req->poll_refs on every io_poll_wake then overflow to 0 which will fput req->file twice and cause a struct file refcount issue. GKE clusters, including Autopilot clusters, with Container-Optimized OS using Linux Kernel version 5.15 are affected. GKE clusters using Ubuntu images or using GKE Sandbox are unaffected.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| Medium | [CVE-CVE-2023-0468](#) | |

GCP-2023-014

Updated: ** 2023-08-11 ****Published:** 2023-06-15

Description

| Description | Severity | Notes | | |

**2023-08-11 Update: **Added patch versions for GKE on VMware, GKE on AWS, GKE on Azure, and Google Distributed Cloud Virtual for Bare Metal.

Two new security issues were discovered in Kubernetes where users may be able to launch containers that bypass policy restrictions when using ephemeral containers and either ImagePolicyWebhook (CVE-2023-2727) or the ServiceAccount admission plugin (CVE-2023-2728).

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| Medium | [CVE-2023-2727](#), [CVE-2023-2728](#) | |

GCP-2023-013

Published: 2023-06-08

Description

| Description | Severity | Notes | | |

When you enable the Cloud Build API in a project, Cloud Build automatically creates a default service account to execute builds on your behalf. This Cloud Build service account previously had the `logging.privateLogEntries.list` IAM permission, which allowed builds to have access to list private logs by default. This permission has now been revoked from the Cloud Build service account to adhere to the security principle of least privilege.

For instructions and more details, see the [Cloud Build security bulletin](#).

| Low | | |

GCP-2023-010

Published: 2023-06-07

Description

| Description | Severity | Notes | | |

Google identified three new vulnerabilities in the gRPC C++ implementation. These will be published soon publicly as [CVE-2023-1428](#), [CVE-2023-32731](#) and [CVE-2023-32732](#).

In April, we identified two vulnerabilities in 1.53 and 1.54 releases. One was a Denial-of-Service vulnerability within the gRPC's C++ implementation and the other was a remote data exfiltration vulnerability. These have been fixed in 1.53.1, 1.54.2 and later releases.

Previously in March, our internal teams discovered a Denial-of-Service vulnerability within the gRPC's C++ implementation while performing routine fuzzing activities. It was found in the gRPC 1.52 release, and has been fixed in the 1.52.2 and 1.53 releases.

What should I do?

Ensure that you're using the latest versions of the following software packages:

- gRPC (C++, Python, Ruby) version 1.52, 1.53, and 1.54 need to upgrade to the following patch releases;
- 1.52.2
- 1.53.1
- 1.54.2
- gRPC (C++, Python, Ruby) version 1.51 and earlier are not affected, so users with these versions can take no action

What vulnerabilities are addressed by these patches?

These patches mitigate the following vulnerabilities:

- 1.53.1, 1.54.2 and later releases address the following: Denial-Of-Service vulnerability in gRPC C++ implementation. Specially crafted requests can cause a termination of connection between a proxy and a backend. Remote data exfiltration vulnerability: Desynchronisation in the HPACK table due to header size limitations can lead to proxy backends leaking header data from other clients connected to a proxy.
- 1.52.2, 1.53, and later releases address the following: Denial-of-Service vulnerability within the gRPC's C++ implementation. Parsing some specifically formed requests can lead to a crash impacting a server.

We recommend upgrading to the latest versions of the following software packages as listed above.

| High (CVE-2023-1428, CVE-2023-32731). Medium (CVE-2023-32732) | [CVE-2023-1428](#), [CVE-2023-32731](#), [CVE-2023-32732](#) | |

GCP-2023-009

Published: 2023-06-06

Description

| Description | Severity | Notes | |

A new vulnerability (CVE-2023-2878) has been discovered in the secrets-store-csi-driver where an actor with access to the driver logs could observe service account tokens.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| None | [CVE-2023-2878](#) | |

GCP-2023-008

Published: 2023-06-05

Description

| Description | Severity | Notes | | |

A new vulnerability (CVE-2023-1872) has been discovered in the Linux kernel that can lead to a privilege escalation to root on the node.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2023-1872](#) | |

GCP-2023-007

Published: 2023-06-02

Description

| Description | Severity | Notes | | |

A vulnerability was recently discovered in Cloud SQL for SQL Server that allowed customer administrator accounts to create triggers in the `tempdb` database and use those to gain `sysadmin` privileges in the instance. The `sysadmin` privileges would give the attacker access to system databases and partial access to the machine running that SQL Server instance.

Google Cloud resolved the issue by patching the security vulnerability by March 1, 2023. Google Cloud didn't find any compromised customer instances.

For instructions and more details, see the [Cloud SQL security bulletin](#).

| High | | |

GCP-2023-005

Published: 2023-05-18

Updated: 2023-06-06

Description

| Description | Severity | Notes | | |

2023-06-06 Update: New GKE versions have been updated to include the latest Ubuntu versions that patch CVE-2023-1281 and CVE-2023-1829.

Two new vulnerabilities (CVE-2023-1281, CVE-2023-1829) have been discovered in the Linux kernel that can lead to a privilege escalation to root on the node.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2023-1281](#) [CVE-2023-1829](#) | |

GCP-2023-004

Published: 2023-04-26

Description

| Description | Severity | Notes | | |

Two vulnerabilities ([CVE-2023-1017](#) and [CVE-2023-1018](#)) were discovered in Trusted Platform Module (TPM) 2.0.

The vulnerabilities could have allowed a sophisticated attacker to exploit a 2-byte out of bounds read/write on certain Compute Engine VMs.

For instructions and more details, see the [Compute Engine security bulletin](#).

| Medium |

- [CVE-2023-1017](#)
- [CVE-2023-1018](#)

| |

GCP-2023-003

Published: 2023-04-11

Updated: 2023-12-21

Description

| Description | Severity | Notes | | |

2023-12-21 Update: Clarify that GKE Autopilot clusters in the default configuration are not impacted.

Two new vulnerabilities, CVE-2023-0240 and CVE-2023-23586, have been discovered in the Linux kernel that could allow an unprivileged user to escalate privileges.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2023-0240](#), [CVE-2023-23586](#) | |

GCP-2023-002

Description

| Description | Severity | Notes | | |

The following CVEs expose Cloud Service Mesh to exploitable vulnerabilities:

- CVE-2023-27496: If Envoy is running with the OAuth filter enabled exposed, a malicious actor could construct a request which would cause denial of service by crashing Envoy.
- CVE-2023-27488: The attacker can use this vulnerability to bypass auth checks when ext_authz is used.
- CVE-2023-27493: Envoy configuration must also include an option to add request headers that were generated using inputs from the request, such as the peer certificate SAN.
- CVE-2023-27492: Attackers can send large request bodies for routes that have Lua filter enabled and trigger crashes.
- CVE-2023-27491: Attackers can send specifically crafted HTTP/2 or HTTP/3 requests to trigger parsing errors on HTTP/1 upstream service.
- CVE-2023-27487: The header `x-envoy-original-path` should be an internal header, but Envoy does not remove this header from the request at the beginning of request processing when it is sent from an untrusted client.

For instructions and more details, see the [Cloud Service Mesh security bulletin](#).

| High |

- [CVE-2023-27496](#)
- [CVE-2023-27488](#)
- [CVE-2023-27493](#)
- [CVE-2023-27492](#)
- [CVE-2023-27491](#)
- [CVE-2023-27487](#)

||

GCP-2023-001

Published: 2023-03-01, 2023-12-21

Description

| Description | Severity | Notes | | |

2023-12-21 Update: Clarify that GKE Autopilot clusters in the default configuration are not impacted.

A new vulnerability (CVE-2022-4696) has been discovered in the Linux kernel that can lead to a privilege escalation on the node.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2022-4696](#) | |

GCP-2022-026

Published: 2023-01-11

Description

| Description | Severity | Notes | | |

Two new vulnerabilities (CVE-2022-3786 and CVE-2022-3602) have been discovered in OpenSSL v3.0.6 that can potentially cause a crash.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| Medium |

- [CVE-2022-3786](#)
- [CVE-2022-3602](#)

||

GCP-2022-025

Published: 2022-12-21 **Updated:** 2023-01-19, 2023-12-21

Description

| Description | Severity | Notes | |

2023-12-21 Update: Clarify that GKE Autopilot clusters in the default configuration are not impacted.

2023-01-19 Update: Added information that GKE version 1.21.14-gke.14100 is available.

Two new vulnerabilities (CVE-2022-3786 and CVE-2022-3602) have been discovered in OpenSSL v3.0.6 that can potentially cause a crash.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| Medium |

- [CVE-2022-3786](#)
- [CVE-2022-3602](#)

||

GCP-2022-024

Published: 2022-11-09

Updated: 2023-01-19

Description

| Description | Severity | Notes | |

2023-01-19 Update: Added information that GKE version 1.21.14-gke.14100 is available.

2022-12-16 Update: Added patch versions for GKE and GKE on VMware.

Two new vulnerabilities (CVE-2022-2585 and CVE-2022-2588) have been discovered in the Linux kernel that can lead to a full container break out to root on the node.

For instructions and more details, see the:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High |

- [CVE-2022-2585](#)
- [CVE-2022-2588](#)

| |

GCP-2022-023

Published: 2022-11-04

Description

| Description | Severity | Notes | |

A security vulnerability, CVE-2022-39278, has been discovered in Istio, which is used in Cloud Service Mesh, that allows a malicious attacker to crash the control plane.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2022-39278](#) | |

GCP-2022-022

Published: 2022-10-28

Updated: 2022-12-14

Description

| Description | Severity | Notes | |

2022-12-14 Update: Added patch versions for GKE and GKE on VMware.

A new vulnerability, CVE-2022-20409, has been discovered in the Linux kernel that could allow an unprivileged user to escalate to system execution privilege.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2022-20409](#) | |

GCP-2022-021

Published: 2022-10-27

Updated: 2023-01-19, 2023-12-21

Description

| Description | Severity | Notes | |

2023-12-21 Update: Clarify that GKE Autopilot clusters in the default configuration are not impacted.

2023-01-19 Update: Added information that GKE version 1.21.14-gke.14100 is available.

2022-12-15 Update: Updated information that version 1.21.14-gke.9400 of Google Kubernetes Engine is pending rollout and may be superseded by a higher version number.

2022-11-22 Update: Added patch versions for GKE on VMware, GKE on AWS, and GKE on Azure.

A new vulnerability, CVE-2022-3176, has been discovered in the Linux kernel that can lead to local privilege escalation. This vulnerability allows an unprivileged user to achieve full container breakout to root on the node.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2022-3176](#) | |

GCP-2022-020

Published: 2022-10-05

Updated: 2022-10-12

Description

| Description | Severity | Notes | | |

The Istio control plane `istiod` is vulnerable to a request processing error, allowing a malicious attacker that sends a specially crafted message which results in the control plane crashing when the validating webhook for a cluster is exposed publicly. This endpoint is served over TLS port 15017, but does not require any authentication from the attacker.

For instructions and more details, see the [Cloud Service Mesh security bulletin](#).

| High | [CVE-2022-39278](#) | |

GCP-2022-019

Published: 2022-09-22

Description

| Description | Severity | Notes | | |

A message parsing and memory management vulnerability in ProtocolBuffer's C++ and Python implementations can trigger an out of memory (OOM) failure when processing a specially crafted message. This could lead to a denial of service (DoS) on services using the libraries.

What should I do?

Ensure that you're using the latest versions of the following software packages:

- `protobuf-cpp` (3.18.3, 3.19.5, 3.20.2, 3.21.6)
- `protobuf-python` (3.18.3, 3.19.5, 3.20.2, 4.21.6)

What vulnerabilities are addressed by this patch?

The patch mitigates the following vulnerability:

A specially constructed small message that causes the running service to allocate large amounts of RAM. The small size of the request means that it is easy to take advantage of the vulnerability and exhaust resources. C++ and Python systems that consume untrusted protobufs would be vulnerable to DoS attacks if they contain a `MessageSet` object in their RPC request.

| Medium | [CVE-2022-1941](#) | |

GCP-2022-018

Published: 2022-08-01

Updated: 2022-09-14, 2023-12-21

Description

| Description | Severity | Notes | |

2023-12-21 Update: Clarify that GKE Autopilot clusters in the default configuration are not impacted.

2022-09-14 Update: Added patch versions for GKE on VMware, GKE on AWS, and GKE on Azure.

A new vulnerability (CVE-2022-2327) has been discovered in the Linux kernel that can lead to local privilege escalation. This vulnerability allows an unprivileged user to achieve a full container breakout to root on the node.

For instructions and more details, see the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

High | [CVE-2022-2327](#) | |

GCP-2022-017

Published: 2022-06-29 **Updated:** 2022-11-22

Description

| Description | Severity | Notes | |

2022-11-22 Update: Workloads using [GKE Sandbox](#) are not affected by these vulnerabilities.

2022-07-21 Update: additional information on GKE on VMware.

A new vulnerability (CVE-2022-1786) has been discovered in the Linux kernel versions 5.10 and 5.11. This vulnerability allows an unprivileged user with local access to the cluster to achieve a full container breakout to root on the node. Only clusters that run Container-Optimized OS are affected. GKE Ubuntu versions use either version 5.4 or 5.15 of the kernel and are not affected.

For instructions and more details, see the:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2022-1786](#) | |

GCP-2022-016

Published: 2022-06-23 **Updated:** 2022-11-22

Description

| Description | Severity | Notes | |

2022-11-22 Update: Autopilot clusters are not affected by CVE-2022-29581 but are vulnerable to CVE-2022-29582 and CVE-2022-1116.

Three new memory corruption vulnerabilities (CVE-2022-29581, CVE-2022-29582, CVE-2022-1116) have been discovered in the Linux kernel. These vulnerabilities allow an unprivileged user with local access to the cluster to achieve a full container breakout to root on the node. All Linux clusters (Container-Optimized OS and Ubuntu) are affected.

For instructions and more details, refer to the following bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High |

- [CVE-2022-29581](#)
- [CVE-2022-29582](#)
- [CVE-2022-1116](#)

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GCP-2022-015

Published: 2022-06-09 **Updated:** 2022-06-10

Description

| Description | Severity | Notes | | |

2022-06-10 Update: The Cloud Service Mesh versions have been updated. For instructions and more details, see the [Cloud Service Mesh security bulletin](#).

The following Envoy and Istio CVEs expose Cloud Service Mesh and Istio on GKE to remotely exploitable vulnerabilities:

- CVE-2022-31045: Istio data plane can potentially access memory unsafely when the Metadata Exchange and Stats extensions are enabled.
- CVE-2022-29225: Data can exceed intermediate buffer limits if a malicious attacker passes a small highly compressed payload (zip bomb attack).
- CVE-2021-29224: Potential null pointer dereference in GrpcHealthCheckerImpl.
- CVE-2021-29226: OAuth filter allows trivial bypass.
- CVE-2022-29228: OAuth filter can corrupt memory (earlier versions) or trigger an ASSERT() (later versions).
- CVE-2022-29227: Internal redirects crash for requests with body or trailers.

For instructions and more details, see the [Cloud Service Mesh security bulletin](#).

| Critical |

- [CVE-2022-31045](#)
- [CVE-2022-29225](#)
- [CVE-2021-29224](#)
- [CVE-2021-29226](#)
- [CVE-2022-29228](#)
- [CVE-2022-29227](#)

||

GCP-2022-014

Published: 2022-04-26 **Updated:** 2022-11-22

Description

| Description | Severity | Notes | | |

2022-11-22 Update: GKE Autopilot clusters and workloads running in [GKE Sandbox](#) are unaffected.

2022-05-12 Update: The GKE on AWS and GKE on Azure versions have been updated. For instructions and more details, see the:

- [GKE on AWS security bulletin](#)
 - [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)
-

Two security vulnerabilities, [CVE-2022-1055](#) and [CVE-2022-27666](#) have been discovered in the Linux kernel. Each can lead to a local attacker being able to perform a container breakout, privilege escalation on the host, or both. These vulnerabilities affect all GKE node operating systems (Container-Optimized OS and Ubuntu). For instructions and more details, see the following security bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2022-1055](#) [CVE-2022-27666](#) | |

GCP-2022-013

Published: 2022-04-11 **Updated:** 2022-04-22

Description

| Description | Severity | Notes | |

A security vulnerability, [CVE-2022-23648](#), has been discovered in containerd's handling of path traversal in the OCI image volume specification. Containers launched through containerd's CRI implementation with a specially-crafted image configuration could gain full read access to arbitrary files and directories on the host. This vulnerability may bypass any policy-based enforcement on container setup (including a Kubernetes Pod Security Policy).

For instructions and more details, see the following security bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| Medium | [CVE-2022-23648](#) | |

GCP-2022-012

Published: 2022-04-07 **Updated:** 2022-11-22

Description

| Description | Severity | Notes | |

2022-11-22 Update: For GKE clusters in both modes, Standard and Autopilot, workloads using [GKE Sandbox](#) are unaffected.

A security vulnerability, CVE-2022-0847, has been discovered in the Linux kernel version 5.8 and later that can potentially escalate container privileges to root. This vulnerability affects the following products:

- GKE node pool versions 1.22 and later that use Container-Optimized OS images (Container-Optimized OS 93 and later)
- GKE on VMware v1.10 for Container-Optimized OS images
- GKE on AWS v1.21 and GKE on AWS (previous generation) v1.19, v1.20, v1.21, which use Ubuntu
- Managed clusters of GKE on Azure v1.21 which use Ubuntu

For instructions and more details, see the following security bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2022-0847](#) | |

GCP-2022-011

Published: 2022-03-22 ****Updated:** **2022-08-11

Description

| Description | Severity | |

Update 2022-08-11: Added more information about the Simultaneous Multi-Threading (SMT) configuration. SMT was intended to be disabled, but was enabled on the versions listed.

If you manually enabled SMT for a sandboxed node pool, SMT will remain manually enabled despite this issue.

There is a misconfiguration with Simultaneous Multi-Threading (SMT), also known as Hyper-threading, on GKE Sandbox images. The misconfiguration leaves nodes potentially exposed to side channel attacks such as Microarchitectural Data Sampling (MDS) (for more context, see [GKE Sandbox documentation](#)). We do not recommend using the following affected versions:

- 1.22.4-gke.1501
- 1.22.6-gke.300
- 1.23.2-gke.300
- 1.23.3-gke.600

For instructions and more details, see the: [GKE security bulletin](#).

| Medium | |

GCP-2022-010

Description

| Description | Severity | Notes | |

The following Istio CVE exposes Cloud Service Mesh to a remotely exploitable vulnerability:

- CVE-2022-24726: The Istio control plane, `istiod`, is vulnerable to a request processing error, allowing a malicious attacker that sends a specially crafted message which results in the control plane crashing when the validating webhook for a cluster is exposed publicly. This endpoint is served over TLS port 15017 but does not require any authentication from the attacker.

For instructions and more details, see the following security bulletin:

- [Cloud Service Mesh security bulletin](#).

| High |

- [CVE-2022-24726](#)

| |

GCP-2022-009

Published: 2022-03-01

Description

| Description | Severity | |

Some unexpected paths to access the node VM on GKE Autopilot clusters could have been used to escalate privileges in the cluster. These issues have been fixed and **no further action is required**. The fixes address issues reported through our [Vulnerability Reward Program](#).

For instructions and more details, see the [GKE security bulletin](#)

| Low | |

GCP-2022-008

Published: 2022-02-23 **Updated:* * 2022-04-28

Description

| Description | Severity | Notes | |

**2022-04-28 Update:* * Added versions of GKE on VMware that fix these vulnerabilities. For details, see the [GKE on VMware security bulletin](#).

The Envoy project recently discovered a set of vulnerabilities. All issues listed below are fixed in Envoy release 1.21.1.

- [CVE-2022-23606](#): When a cluster is deleted via Cluster Discovery Service (CDS) all idle connections established to endpoints in that cluster are disconnected. A recursion was erroneously introduced in Envoy version 1.19 to the procedure of disconnecting idle connections that can lead to stack exhaustion and abnormal process termination when a cluster has a large number of idle connections.
- [CVE-2022-21655](#): Envoy's internal redirect code assumes that a route entry exists. When an internal redirect is done to a route which has a direct response entry and no route entry, it results in dereferencing a null pointer and crashing.
- [CVE-2021-43826](#): When Envoy is configured to use tcp_proxy which uses upstream tunneling (over HTTP), and downstream TLS termination, Envoy will crash if the downstream client disconnects during the TLS handshake while the upstream HTTP stream is still being established. The downstream disconnect can be either client or server initiated. The client can disconnect for any reason. The server may disconnect if, for example, it has no TLS ciphers or TLS protocol versions compatible with the client. It may be possible to trigger this crash in other downstream configurations as well.
- [CVE-2021-43825](#): Sending a locally generated response must stop further processing of request or response data. Envoy tracks the amount of buffered request and response data and aborts the request if the amount of buffered data is over the limit by sending 413 or 500 responses. However when locally generated response is sent because of the internal buffer overflows while response is processed by the filter chain the operation may not be aborted correctly and result in accessing a freed memory block.
- [CVE-2021-43824](#): Envoy crashes when using the JWT filter with a "safe_regex" match rule and a specially crafted request like "CONNECT host:port HTTP/1.1". When reaching the JWT filter, a "safe_regex" rule should evaluate the URL path but there is none here, and Envoy crashes with segfaults.
- [CVE-2022-21654](#): Envoy would incorrectly allow TLS session resumption after mTLS validation settings had been reconfigured. If a client certificate was allowed with the old

configuration but disallowed with the new configuration, the client could resume the previous TLS session even though the current configuration should disallow it. Changes to the following settings are affected:

- `match_subject_alt_names`
- CRL changes
- `allow_expired_certificate`
- `Trust_chain_verification`
- `only_verify_leaf_cert_crl`
- [CVE-2022-21657](#): Envoy does not restrict the set of certificates it accepts from the peer, either as a TLS client or a TLS server, to only those certificates that contain the necessary `extendedKeyUsage` (`id-kp-serverAuth` and `id-kp-clientAuth`, respectively). This means that a peer may present an e-mail certificate (e.g. `id-kp-emailProtection`), either as a leaf certificate or as a CA in the chain, and it will be accepted for TLS. This is particularly bad when combined with [CVE-2022-21656](#), in that it allows a Web PKI CA that is intended only for use with S/MIME, and thus exempted from audit or supervision, to issue TLS certificates that will be accepted by Envoy.
- [CVE-2022-21656](#): The validator implementation used to implement the default certificate validation routines has a "type confusion" bug when processing `subjectAltNames`. This processing allows, for example, an `rfc822Name` or `uniformResourceIndicator` to be authenticated as a domain name. This confusion allows for the bypassing of `nameConstraints`, as processed by the underlying OpenSSL/BoringSSL implementation, exposing the possibility of impersonation of arbitrary servers.

For detailed instructions regarding specific products, see the following security bulletins:

- [Cloud Service Mesh security bulletin](#)
- [Istio on GKE security bulletin](#)
- [GKE](#)
- [GKE on VMware](#)
- [Google Distributed Cloud Virtual for Bare Metal](#)

What should I do? Envoy users managing their own Envoys should ensure that they are using Envoy release 1.21.1. Envoy users managing their own Envoys build the binaries from a source like GitHub and deploy them.

There's no action to be taken by users who run managed Envoys (Google Cloud provides the Envoy binaries), for which Google Cloud products will switch to 1.21.1. | High | [CVE-2022-23606](#) [CVE-2022-21655](#) [CVE-2021-43826](#) [CVE-2021-43825](#) [CVE-2021-43824](#) [CVE-2022-21654](#) [CVE-2022-21657](#) [CVE-2022-21656](#) | |

GCP-2022-007

Published: 2022-02-22

Description

| Description | Severity | Notes | |

The following Envoy and Istio CVEs expose Cloud Service Mesh and Istio on GKE to remotely exploitable vulnerabilities:

- CVE-2022-23635: Istiod crashes upon receiving requests with a specially crafted `authorization` header.
- CVE-2021-43824: Potential null pointer dereference when using JWT filter `safe_regex` match
- CVE-2021-43825: Use-after-free when response filters increase response data, and increased data exceeds downstream buffer limits.
- CVE-2021-43826: Use-after-free when tunneling TCP over HTTP, if downstream disconnects during upstream connection establishment.
- CVE-2022-21654: Incorrect configuration handling allows mTLS session re-use without re-validation after validation settings have changed.
- CVE-2022-21655: Incorrect handling of internal redirects to routes with a direct response entry.
- CVE-2022-23606: Stack exhaustion when a cluster is deleted via Cluster Discovery Service.

For instructions and more details, see the following security bulletins:

- [Cloud Service Mesh security bulletin](#).
- [Istio on GKE security bulletin](#).

| High |

- [CVE-2022-23635](#)
- [CVE-2021-43824](#)
- [CVE-2021-43825](#)
- [CVE-2021-43826](#)
- [CVE-2022-21654](#)
- [CVE-2022-21655](#)
- [CVE-2022-23606](#)

| |

GCP-2022-006

Published: 2022-02-14 ****Updated:** **2022-05-16

Description

| Description | Severity | Notes | |

2022-05-16 Update: Added GKE version 1.19.16-gke.7800 or later to the list of versions that have code to fix this vulnerability. For details, see the [GKE security bulletin](#).

2022-05-12 Update: The GKE, GKE on VMware, GKE on AWS, and GKE on Azure versions have been updated. For instructions and more details, see the:

- [GKE security bulletin](#)
 - [GKE on VMware security bulletin](#)
 - [GKE on AWS security bulletin](#)
 - [GKE on Azure security bulletin](#)
-

A security vulnerability, CVE-2022-0492, has been discovered in the Linux kernel's `cgroup_release_agent_write` function. The attack uses unprivileged user namespaces and under certain circumstances this vulnerability can be exploitable for container breakout.

| Low |

For instructions and more details, see the:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [GKE on Azure security bulletin](#)

||

GCP-2022-005

Published: 2022-02-11 ****Updated:** **2022-02-15

Description

| Description | Severity | Notes | | |

A security vulnerability, CVE-2021-43527, has been discovered in any binary that links to the vulnerable versions of libnss3 found in NSS (Network Security Services) versions prior to 3.73 or 3.68.1. Applications using NSS for certificate validation or other TLS, X.509, OCSP or CRL functionality may be impacted, depending on how NSS is used/configured.

For instructions and more details, see the:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on Azure security bulletin](#)

| Medium | [CVE-2021-43527](#) | |

GCP-2022-004

Published: 2022-02-04

Description

| Description | Severity | Notes | |

A security vulnerability, [CVE-2021-4034](#), has been discovered in pkexec, a part of the Linux policy kit package (polkit), that allows an authenticated user to perform a privilege escalation attack. PolicyKit is generally used only on Linux desktop systems to allow non-root users to perform actions such as rebooting the system, installing packages, restarting services etc, as governed by a policy.

For instructions and more details, see the:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on Azure security bulletin](#)

| None | [CVE-2021-4034](#) | |

GCP-2022-002

Published: 2022-02-01 **Updated:** 2022-02-25

Description

| Description | Severity | Notes | |

2022-02-25 Update: The GKE versions have been updated. For instructions and more details, see the:

- [GKE security bulletin](#)

**2022-02-23 Update:* * The GKE and GKE on VMware versions have been updated. For instructions and more details, see the:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)

| |

2022-02-04 Update: The rollout start date for GKE patch versions was February 2.

Three security vulnerabilities, [CVE-2021-4154](#), [CVE-2021-22600](#), and [CVE-2022-0185](#), have been discovered in the Linux kernel, each of which can lead to either a container breakout, privilege escalation on the host, or both. These vulnerabilities affect all node operating systems (COS and Ubuntu) on GKE, GKE on VMware, GKE on AWS (current and previous generation), and GKE on Azure. Pods using [GKE Sandbox](#) are not vulnerable to these vulnerabilities. See the [COS release notes](#) for more details.

For instructions and more details, see the:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)

High |

- [CVE-2021-4154](#)
- [CVE-2021-22600](#)
- [CVE-2022-0185](#)

|

GCP-2022-001

Published: 2022-01-06

Description

| Description | Severity | Notes | |

A potential Denial of Service issue in `protobuf-java` was discovered in the parsing procedure for binary data.

What should I do?

Ensure that you're using the latest versions of the following software packages:

- `protobuf-java` (3.16.1, 3.18.2, 3.19.2)
- `protobuf-kotlin` (3.18.2, 3.19.2)
- `google-protobuf` [JRuby gem] (3.19.2)

Protobuf "javalite" users (typically Android) are not affected.

**What vulnerabilities are addressed by this patch? **

The patch mitigates the following vulnerability:

An implementation weakness in how unknown fields are parsed in Java. A small (~800 KB) malicious payload can occupy the parser for several minutes by creating large numbers of short-lived objects that cause frequent, repeated garbage collection pauses.

| High | [CVE-2021-22569](#) | |

GCP-2021-024

Published: 2021-10-21

Description

| Description | Severity | Notes | |

A security issue was discovered in the Kubernetes **ingress-nginx** controller, [CVE-2021-25742](#). Ingress-nginx custom snippets allows retrieval of ingress-nginx service account

tokens and secrets across all namespaces.

For instructions and more details, see the:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

None | [CVE-2021-25742](#) | | |

GCP-2021-019

Published: 2021-09-29

Description

| Description | Severity | Notes | | |

There is a known issue where updating a `BackendConfig` resource using the `v1beta1` API removes an active Google Cloud Armor security policy from its service.

For instructions and more details, see the [GKE security bulletin](#).

| Low | | |

GCP-2021-022

Published: 2021-09-22

Description

| Description | Severity | Notes | | |

A vulnerability has been discovered in the GKE Enterprise Identity Service (AIS) LDAP module of GKE on VMware versions 1.8 and 1.8.1 where a seed key used in generating keys is predictable. With this vulnerability, an authenticated user could add arbitrary claims and escalate privileges indefinitely.

For instructions and more details, see the [GKE on VMware security bulletin](#).

| High | | |

GCP-2021-021

Published: 2021-09-22

Description

| Description | Severity | Notes | | |

A security vulnerability, CVE-2020-8561, has been discovered in Kubernetes where certain webhooks can be made to redirect kube-apiserver requests to private networks of that API server.

For instructions and more details, see the:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| Medium | [CVE-2020-8561](#) | |

GCP-2021-023

Published: 2021-09-21

Description

| Description | Severity | Notes | |

Per VMware security advisory VMSA-2021-0020, VMware received reports of multiple vulnerabilities in vCenter. VMware has made updates available to remediate these vulnerabilities in affected VMware products.

We have already applied the patches provided by VMware for the vSphere stack to Google Cloud VMware Engine per the VMware security advisory. This update addresses the security vulnerabilities described in CVE-2021-22005, CVE-2021-22006, CVE-2021-22007, CVE-2021-22008, and CVE-2021-22010. Other non-critical security issues will be addressed in the upcoming VMware stack upgrade (per the advance notice sent in July, more details will be provided soon on the specific timeline of the upgrade).

VMware Engine impact

Based on our investigations, no customers were found to be impacted.

What should I do?

Because VMware Engine clusters are not affected by this vulnerability, no further action is required.

| Critical |

- [VMSA-2021-0020](#)
- [CVE-2021-22005](#)
- [CVE-2021-22006](#)
- [CVE-2021-22007](#)
- [CVE-2021-22008](#)
- [CVE-2021-22010](#)

| |

GCP-2021-020

Published: 2021-09-17

Description

| Description | Severity | Notes | |

Certain Google Cloud load balancers routing to an Identity-Aware Proxy (IAP) enabled [Backend Service](#) could have been vulnerable to an untrusted party under limited conditions. This addresses an issue reported through our Vulnerability Reward Program.

The conditions were that the servers:

- Were HTTP(S) load balancers **and**
- Used a default backend or a backend that had a wildcard host mapping rule (that is, `host="*"`)

In addition, a user in your organization must have clicked a specifically-crafted link sent by an untrusted party.

This issue has now been resolved. IAP has been updated to issue cookies only to authorized hosts as of September 17, 2021. A host is considered authorized if it matches at least one Subject Alternative Name (SAN) in one of the certificates installed on your load balancers.

What to do

Some of your users may experience an HTTP 401 Unauthorized response with an IAP error code 52 while trying to access apps or services. This error code means that the client sent a `Host` header which does not match any Subject Alternative Names associated with the load balancer's SSL certificate(s). The load balancer administrator needs to update the SSL certificate to ensure that the Subject Alternative Name (SAN) list contains all the hostnames through which users are accessing the IAP-protected apps or services. Learn more about [IAP error codes](#).

| High | | |

GCP-2021-018

Published: 2021-09-15 **Updated:** 2021-09-20

Description

| Description | Severity | Notes | |

A security issue was discovered in Kubernetes, CVE-2021-25741, where a user may be able to create a container with subpath volume mounts to access files & directories outside of the volume, including on the host filesystem.

For instructions and more details, see the:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

| High | [CVE-2021-25741](#) | |

GCP-2021-017

Published: 2021-09-01 **Updated:** 2021-09-23

Description

| Description | Severity | Notes | | |

****2021-09-23 update:** **Containers running inside of [GKE Sandbox](#) are unaffected by this vulnerability for attacks originating inside the container.

Two security vulnerabilities, CVE-2021-33909 and CVE-2021-33910, have been discovered in the Linux kernel that can lead to an OS crash or an escalation to root by an unprivileged user. This vulnerability affects all GKE node operating systems (COS and Ubuntu).

For instructions and more details, see the following security bulletins:

- [GKE security bulletin](#).
- [GKE on AWS security bulletin](#).

| High | [CVE-2021-33909](#), [CVE-2021-33910](#) | |

GCP-2021-016

Published: 2021-08-24

Description

| Description | Severity | Notes | | |

The following Envoy and Istio CVEs expose Cloud Service Mesh and Istio on GKE to remotely exploitable vulnerabilities:

- CVE-2021-39156: HTTP requests with a fragment (a section in the end of a URI that begins with a `#` character) in the URI path could bypass Istio's URI path-based authorization policies.
- CVE-2021-39155: HTTP requests could potentially bypass an Istio authorization policy when using rules based on `hosts` or `notHosts`.
- CVE-2021-32781: Affects Envoy's `decompressor`, `json-transcoder`, or `grpc-web` extensions or proprietary extensions that modify and increase the size of request or response bodies. Modifying and increasing the size of the body in an Envoy's extension

beyond the internal buffer size could lead to Envoy accessing deallocated memory and terminating abnormally.

- CVE-2021-32780: An untrusted upstream service could cause Envoy to terminate abnormally by sending the `GOAWAY` frame followed by the `SETTINGS` frame with the `SETTINGS_MAX_CONCURRENT_STREAMS` parameter set to `0`. *(Not applicable to Istio on GKE)*
- CVE-2021-32778: An Envoy client opening and then resetting a large number of HTTP/2 requests could lead to excessive CPU consumption. *(Not applicable to Istio on GKE)*
- CVE-2021-32777: HTTP requests with multiple value headers could do an incomplete authorization policy check when the `ext_authz` extension is used.

For instructions and more details, see the following security bulletins:

- [Cloud Service Mesh security bulletin](#).
- [Istio on GKE security bulletin](#).

| High |

- [CVE-2021-39156](#)
- [CVE-2021-39155](#)
- [CVE-2021-32781](#)
- [CVE-2021-32780](#)
- [CVE-2021-32778](#)
- [CVE-2021-32777](#)

||

GCP-2021-015

Published: 2021-07-13 **Updated:** 2021-07-15

Description

| Description | Severity | Notes | |

A new security vulnerability, [CVE-2021-22555](#), has been discovered where a malicious actor with `CAP_NET_ADMIN` privileges can potentially cause a container breakout to root on the host. This vulnerability affects all GKE clusters and GKE on VMware running Linux version 2.6.19 or later.

For instructions and more details, see the following security bulletins:

- [GKE security bulletin](#).
- [GKE on VMware security bulletin](#).

| High | [CVE-2021-22555](#) | |

GCP-2021-014

Published: 2021-07-05

Description

| Description | Severity | Notes | | |

Microsoft published a security bulletin on a Remote code execution (RCE) vulnerability, CVE-2021-34527, that affects the print spooler in Windows servers. The CERT Coordination Center (CERT/CC) published an update note on a related vulnerability, dubbed "PrintNightmare" that also affects Windows print spoolers - [PrintNightmare, Critical Windows Print Spooler Vulnerability](#)

For instructions and more details, see the [GKE security bulletin](#).

| High | [CVE-2021-34527](#) | |

GCP-2021-012

Published: 2021-06-24 **Updated:** 2021-07-09

Description

| Description | Severity | Notes | | |

The Istio project recently [announced](#) a security vulnerability where credentials specified in the Gateway and DestinationRule credentialName field can be accessed from different namespaces.

For product-specific instructions and more details, see:

- [Cloud Service Mesh security bulletin](#).
- [GKE Enterprise security bulletin GCP-2021-012](#) for information specific to Google Kubernetes Engine, Google Distributed Cloud Virtual for Bare Metal, and GKE on VMware.

| High | [CVE-2021-34824](#) | |

GCP-2021-011

Published: 2021-06-04 **Updated:** 2021-10-19

Description

| Description | Severity | Notes | | |

**2021-10-19 update: **

For instructions and more details, see the following security bulletins:

- [GKE on VMware security bulletin](#)
 - [GKE on AWS security bulletin](#)
 - [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)
-

The security community recently disclosed a new security vulnerability ([CVE-2021-30465](#)) found in `runc` that has the potential to allow full access to a node filesystem.

For GKE, because exploiting this vulnerability requires the ability to create pods, we have rated the severity of this vulnerability at MEDIUM.

For instructions and more details, see the [GKE security bulletin](#).

| Medium | [CVE-2021-30465](#) | |

GCP-2021-010

Published: 2021-05-25

Description

| Description | Severity | Notes | | |

Per VMware security advisory VMSA-2021-0010, remote code execution and authentication bypass vulnerabilities in vSphere Client (HTML5) were privately reported to VMware. VMware has made updates available to remediate these vulnerabilities in affected VMware products.

We have applied the patches provided by VMware for the vSphere stack per the VMware security advisory. This update addresses security vulnerabilities described in CVE-2021-21985 and CVE-2021-21986. The image versions running in your VMware Engine private cloud do not reflect any change at this time to indicate the patches applied. Please rest assured that appropriate patches have been installed and your environment is secured from these vulnerabilities.

VMware Engine impact

Based on our investigations, no customers were found to be impacted.

What should I do?

Because VMware Engine clusters are not affected by this vulnerability, no further action is required.

| Critical |

- [VMSA-2021-0010](#)
- [CVE-2021-21985](#)
- [CVE-2021-21986](#)

| |

GCP-2021-008

Published: 2021-05-17

Description

| Description | Severity | Notes | | |

Istio contains a remotely exploitable vulnerability where an external client can access unexpected services in the cluster, bypassing authorization checks, when a gateway is configured with `AUTO_PASSTHROUGH` routing configuration.

For instructions and more details, see the [Cloud Service Mesh security bulletin](#).

|

High

| [CVE-2021-31921](#) | |

GCP-2021-007

Published: 2021-05-17

Description

| Description | Severity | Notes | | |

Istio contains a remotely exploitable vulnerability where an HTTP request path with multiple slashes or escaped slash characters (`%2F` or `%5C`) could potentially bypass an Istio authorization policy when path based authorization rules are used.

For instructions and more details, see the [Cloud Service Mesh security bulletin](#).

|

High

| [CVE-2021-31920](#) | |

GCP-2021-006

Published: 2021-05-11

Description

| Description | Severity | Notes | | |

The Istio project recently [disclosed](#) a new security vulnerability ([CVE-2021-31920](#)) affecting Istio.

Istio contains a remotely-exploitable vulnerability where an HTTP request with multiple slashes or escaped slash characters can bypass Istio authorization policy when path based authorization rules are used.

For instructions and more details, see the:

- [GKE security bulletin](#)

|

High

| [CVE-2021-31920](#) | |

GCP-2021-005

Published: 2021-05-11

Description

| Description | Severity | Notes | |

A reported vulnerability has shown that Envoy does not decode escaped slash sequences `%2F` and `%5C` in HTTP URL paths in Envoy versions 1.18.2 and earlier. In addition, some Envoy-based products do not enable path normalization controls. A remote attacker may craft a path with escaped slashes (for example, `/something%2F..%2Fadmin`), to bypass access control (for example, a block on `/admin`). A backend server could then decode slash sequences and normalize the path to provide an attacker access beyond the scope provided for by the access control policy.

What should I do?

If backend servers treat `/` and `%2F` or `\` and `%5C` interchangeably and a URL path based matching is configured, we recommend reconfiguring the backend server to not treat `\` and `%2F` or `\` and `%5C` interchangeably, if feasible.

What behavioral changes were introduced?

Envoy's [normalize_path](#) and [merge adjacent slashes](#) options were enabled to address other common path confusion vulnerabilities in Envoy-based products.

|

High

| [CVE-2021-29492](#) | |

GCP-2021-004

Published: 2021-05-06

Description

| Description | Severity | Notes | |

The Envoy and Istio projects recently [announced](#) several new security vulnerabilities ([CVE-2021-28683](#), [CVE-2021-28682](#) and [CVE-2021-29258](#)), that could allow an attacker to crash Envoy.

Google Kubernetes Engine clusters do not run Istio by default and are not vulnerable. If Istio has been installed in a cluster and configured to expose services to the internet, those services may be vulnerable to denial of service.

Google Distributed Cloud Virtual for Bare Metal and GKE on VMware use Envoy by default for Ingress, so Ingress services may be vulnerable to denial of service.

For instructions and more details, see the following security bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

|

Medium

|

- [CVE-2021-28683](#)
- [CVE-2021-28682](#)
- [CVE-2021-29258](#)

||

GCP-2021-003

Published: 2021-04-19

Description

| Description | Severity | Notes | |

The Kubernetes project recently [announced](#) a new security vulnerability, [CVE-2021-25735](#), that could allow node updates to bypass a Validating Admission Webhook.

In a scenario where an attacker has sufficient privileges and where a Validating Admission Webhook is implemented that uses old `Node` object properties (for example fields in `Node.NodeSpec`), the attacker could update properties of a node that could lead to a cluster compromise. None of the policies enforced by GKE and Kubernetes built-in admission controllers are affected, but we recommend customers check any additional admission webhooks they have installed.

For instructions and more details, see the following security bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)

|

Medium

|

[CVE-2021-25735](#)

||

GCP-2021-002

Published: 2021-03-05

Description

| Description | Severity | Notes | | |

Per VMware security advisory VMSA-2021-0002, VMware received reports of multiple vulnerabilities in VMware ESXi and vSphere Client (HTML5). VMware has made updates available to remediate these vulnerabilities in affected VMware products.

We have applied the officially documented workarounds for the vSphere stack per the VMware security advisory. This update addresses security vulnerabilities described in CVE-2021-21972, CVE-2021-21973, and CVE-2021-21974.

VMware Engine impact

Based on our investigations, no customers were found to be impacted.

What should I do?

Because VMware Engine clusters are not affected by this vulnerability, no further action is required.

| Critical |

- [VMSA-2021-0002](#)
- [CVE-2021-21972](#)
- [CVE-2021-21973](#)
- [CVE-2021-21974](#)

||

GCP-2021-001

Published: 2021-01-28

Description

| Description | Severity | Notes | | |

A vulnerability was recently discovered in the Linux utility `sudo`, described in [CVE-2021-3156](#), that may allow an attacker with unprivileged local shell access on a system with `sudo` installed to escalate their privileges to root on the system.

The underlying infrastructure that runs Compute Engine is not impacted by this vulnerability.

All Google Kubernetes Engine (GKE), GKE on VMware, GKE on AWS, and Google Distributed Cloud Virtual for Bare Metal clusters are not affected by this vulnerability.

For instructions and more details, see the following security bulletins:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)
- [Google Distributed Cloud Virtual for Bare Metal security bulletin](#)
- [Compute Engine security bulletin](#)

| None | [CVE-2021-3156](#) | |

GCP-2020-015

Published: 2020-12-07 **Updated:** 2020-12-22

Description

| Description | Severity | Notes | |

Updated: 2021-12-22 The command for GKE in the following section should use `gcloud beta` instead of the `gcloud` command.

```
gcloud container clusters update --no-enable-service-externalips
```

Updated: 2021-12-15 For GKE, the following mitigation is now available:

- Starting in GKE version 1.21, services with ExternalIPs are blocked by a `DenyServiceExternalIPs` admission controller that is enabled by default for new clusters.
- Customers who upgrade to GKE version 1.21 can block services with ExternalIPs using the following command:

```
gcloud container clusters update --no-enable-service-externalips
```

For more information, see [Hardening your cluster's security](#).

The Kubernetes project [recently discovered](#) a new security vulnerability, [CVE-2020-8554](#), that might allow an attacker who has obtained permissions to create a Kubernetes Service of type LoadBalancer or ClusterIP to intercept network traffic originating from other Pods in the cluster. This vulnerability by itself does not give an attacker permissions to create a Kubernetes Service.

All Google Kubernetes Engine (GKE), GKE on VMware, and GKE on AWS clusters are affected by this vulnerability.

What should I do?

For instructions and more details, see the:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)

|

Medium

| [CVE-2020-8554](#) | |

GCP-2020-014

Published: 2020-10-20 **Updated:** 2020-10-20

Description

| Description | Severity | Notes | | |

The Kubernetes project recently discovered [several issues](#) that allow for the exposure of secret data when verbose logging options are enabled. The issues are:

- CVE-2020-8563: Secret leaks in logs for vSphere Provider kube-controller-manager
- CVE-2020-8564: Docker config secrets leaked when file is malformed and loglevel ≥ 4
- CVE-2020-8565: Incomplete fix for CVE-2019-11250 in Kubernetes allows for token leak in logs when logLevel ≥ 9 . Discovered by GKE Security.
- CVE-2020-8566: Ceph RBD adminSecrets exposed in logs when loglevel ≥ 4

What should I do?

No further action is required due to the default verbosity logging levels of GKE.

|

None

|

- [CVE-2020-8563](#)
- [CVE-2020-8564](#)
- [CVE-2020-8565](#)
- [CVE-2020-8566](#)

| |

Google Cloud impact

Per-product details are listed below.

|
Product

|
Impact

|||

Google Kubernetes Engine (GKE)

|
Google Kubernetes Engine (GKE) is not affected.

|||

GKE On-Prem

|
GKE On-Prem is not affected.

|||

GKE on AWS

|
GKE on AWS is not affected.

||

GCP-2020-013

Published: 2020-09-29

Description

Microsoft has disclosed the following vulnerability:

|
Vulnerability

|
Severity

|
CVE

|||

CVE-2020-1472 — A vulnerability in Windows Server allows attackers to use Netlogon Remote Protocol to run a specially-crafted application on a device on the network.

|

NVD Base Score: 10 (Critical)

|

[CVE-2020-1472](#)

||

For more information, see the [Microsoft disclosure](#).

Google Cloud impact

The infrastructure hosting the Google Cloud and Google products is not impacted by this vulnerability. Additional per-product details are listed below.

|

Product

|

Impact

|||

Compute Engine

|

[CVE-2020-1472](#)

For most customers, no further action is required.

Customers using Compute Engine virtual machines running Windows Server should ensure their instances have been updated with the latest Windows [patch](#) or use Windows Server images published after 08-17-2020 (v20200813 or higher).

|||

Google Kubernetes Engine

|

[CVE-2020-1472](#)

For most customers, no further action is required.

Any customers hosting domain controllers in their GKE Windows Server nodes should ensure both the nodes and the containerized workloads that run on those nodes have the latest Windows node image when it is available. A new node image version will be announced in the [GKE release notes](#) in October.

|||

Managed Service for Microsoft Active Directory

|

[CVE-2020-1472](#)

For most customers, no further action is required.

The August patch released by Microsoft that includes fixes to the NetLogon protocol has been applied to all Managed Microsoft AD domain controllers. This patch delivers functionality to protect against potential exploitation. The timely application of patches is one of the key advantages of using the Managed Service for Microsoft Active Directory. Any customers manually running Microsoft Active Directory (and not utilizing Google Cloud's managed service) should ensure their instances have the latest Windows patch or use Windows Server images.

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Google Workspace

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No customer action is required.

This service is not impacted by this vulnerability.

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App Engine standard environment

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No customer action is required.

This service is not impacted by this vulnerability.

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App Engine flexible environment

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No customer action is required.

This service is not impacted by this vulnerability.

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Cloud Run

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No customer action is required.

This service is not impacted by this vulnerability.

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Cloud Run functions

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No customer action is required.

This service is not impacted by this vulnerability.

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Managed Service for Apache Airflow

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No customer action is required.

This service is not impacted by this vulnerability.

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Dataflow

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No customer action is required.

This service is not impacted by this vulnerability.

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Managed Service for Apache Spark

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No customer action is required.

This service is not impacted by this vulnerability.

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Cloud SQL

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No customer action is required.

This service is not impacted by this vulnerability.

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GCP-2020-012

****Published:** **2020-09-14 ****Updated:** **2020-09-17

Description

| Description | Severity | Notes | |

A vulnerability was recently discovered in the Linux kernel, described in [CVE-2020-14386](#), that may allow container escape to obtain root privileges on the host node.

All GKE nodes are affected. Pods running in [GKE Sandbox](#) are not able to leverage this vulnerability.

For instructions and more details, see the:

- [GKE security bulletin.](#)
- [GKE on VMware security bulletin.](#)
- [GKE on AWS security bulletin.](#)

What vulnerability is addressed by this patch?

The patch mitigates the following vulnerability:

The vulnerability [CVE-2020-14386](#), which allows containers with CAP_NET_RAW to write 1 to 10 bytes of kernel memory, and possibly escape the container and obtain root privileges on the host node. This is rated as a High severity vulnerability. |

High

|

[CVE-2020-14386](#)

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GCP-2020-011

Published: 2020-07-24

Description

| Description | Severity | Notes | |

A networking vulnerability, [CVE-2020-8558](#), was recently discovered in Kubernetes. Services sometimes communicate with other applications running inside the same Pod using the local loopback interface (127.0.0.1). This vulnerability allows an attacker with access to the cluster's network to send traffic to the loopback interface of adjacent Pods and nodes. Services that rely on the loopback interface not being accessible outside their Pod could be exploited.

For instructions and more details, see the:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)

|

Low (GKE and GKE on AWS), Medium (GKE on VMware)

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[CVE-2020-8558](#)

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GCP-2020-010

Published: 2020-07-27

Description

Microsoft has disclosed the following vulnerability:

|

Vulnerability

|
Severity

|
CVE

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CVE-2020-1350 — Windows Servers that serve in a DNS server capacity can be exploited to run untrusted code by the Local System Account.

|
NVD Base Score: 10.0 (Critical)

|
[CVE-2020-1350](#)

||
For more information, see the [Microsoft disclosure](#).

Google Cloud impact

The infrastructure hosting the Google Cloud and Google products is not impacted by this vulnerability. Additional per-product details are listed below.

|
Product

|
Impact

|||

Compute Engine

|
[CVE-2020-1350](#)

For most customers, no further action is required.

Customers using Compute Engine virtual machines running Windows Server in a DNS server capacity should ensure their instances have the latest Windows patch or use Windows Server images provided since 07/14/2020.

|||

Google Kubernetes Engine

|
[CVE-2020-1350](#)

For most customers, no further action is required.

Customers using GKE with Windows Server node in a DNS server capacity must manually [update](#) the nodes and the containerized workloads that run on those nodes to a Windows server version containing the fix.

|||

Managed Service for Microsoft Active Directory

|

[CVE-2020-1350](#)

For most customers, no further action is required.

All Managed Microsoft AD domains have been automatically updated with the patched image. Any customers manually running Microsoft Active Directory (and not utilizing Managed Microsoft AD) should ensure their instances have the latest Windows patch or use Windows Server images provided since 07/14/2020.

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Google Workspace

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No customer action is required.

This service is not impacted by this vulnerability.

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App Engine standard environment

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No customer action is required.

This service is not impacted by this vulnerability.

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App Engine flexible environment

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No customer action is required.

This service is not impacted by this vulnerability.

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Cloud Run

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No customer action is required.

This service is not impacted by this vulnerability.

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Cloud Run functions

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No customer action is required.

This service is not impacted by this vulnerability.

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Managed Service for Apache Airflow

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No customer action is required.

This service is not impacted by this vulnerability.

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Dataflow

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No customer action is required.

This service is not impacted by this vulnerability.

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Managed Service for Apache Spark

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No customer action is required.

This service is not impacted by this vulnerability.

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Cloud SQL

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No customer action is required.

This service is not impacted by this vulnerability.

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GCP-2020-009

Published: 2020-07-15

Description

| Description | Severity | Notes |||

A privilege escalation vulnerability, [CVE-2020-8559](#), was recently discovered in Kubernetes. This vulnerability allows an attacker that has already compromised a node to execute a command in any Pod in the cluster. The attacker can thereby use the already compromised node to compromise other nodes and potentially read information, or cause destructive actions.

Note that for an attacker to exploit this vulnerability, a node in your cluster must have already been compromised. This vulnerability, by itself, will not compromise any nodes in your cluster.

For instructions and more details, see the:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)

|

Medium

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[CVE-2020-8559](#)

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GCP-2020-008

Published: 2020-06-19

Description

| Description | Severity | Notes | | |

Description

VMs that have OS Login enabled might be susceptible to privilege escalation vulnerabilities. These vulnerabilities gives users that are granted OS Login permissions (but not given admin access) the ability to escalate to root access in the VM.

For instructions and more details, see the [Compute Engine security bulletin](#).

| High |

- [CVE-2020-8903](#)
- [CVE-2020-8907](#)
- [CVE-2020-8933](#)

||

GCP-2020-007

Published: 2020-06-01

Description

| Description | Severity | Notes | | |

Server Side Request Forgery (SSRF) vulnerability, [CVE-2020-8555](#), was recently discovered in Kubernetes, allowing certain authorized users to leak up to 500 bytes of sensitive information from the control plane host network. The Google Kubernetes Engine (GKE) control plane uses controllers from Kubernetes and is thus affected by this vulnerability. We recommend that you upgrade the control plane to the latest patch version. A node upgrade is not required.

For instructions and more details, see the:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)

|

Medium

|

[CVE-2020-8555](#)

||

GCP-2020-006

Published: 2020-06-01

Description

| Description | Severity | Notes | |

Kubernetes has disclosed a [vulnerability](#) that allows a privileged container to redirect node traffic to another container. Mutual TLS/SSH traffic, such as between the kubelet and API server or traffic from applications using mTLS cannot be read or modified by this attack. All Google Kubernetes Engine (GKE) nodes are affected by this vulnerability, and we recommend that you upgrade to the latest patch version.

For instructions and more details, see the:

- [GKE security bulletin](#)
- [GKE on VMware security bulletin](#)
- [GKE on AWS security bulletin](#)

|

Medium

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[Kubernetes issue 91507](#)

||

GCP-2020-005

Published: 2020-05-07

Description

|
Vulnerability

|
Severity

|
CVE

|||

A vulnerability was recently discovered in the Linux kernel, described in [CVE-2020-8835](#), allowing container escape to obtain root privileges on the host node.

Google Kubernetes Engine (GKE) Ubuntu nodes running GKE 1.16 or 1.17 are affected by this vulnerability, and we recommend that you upgrade to the latest patch version as soon as possible.

Please see the [GKE security bulletin](#) for instructions and more details.

|
High

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[CVE-2020-8835](#)

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GCP-2020-004

****Published:** **2020-03-31 ****Updated:** **2020-03-31

Description

Kubernetes has disclosed the following vulnerabilities:

|
Vulnerability

|
Severity

|
CVE

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CVE-2019-11254 — This is a Denial of Service (DoS) vulnerability that impacts the API server.

|

Medium

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[CVE-2019-11254](#)

||

See the [GKE on VMware security bulletin](#) for instructions and more details.

GCP-2020-003

**Published: **2020-03-31 **Updated: **2020-03-31

Description

Kubernetes has disclosed the following vulnerabilities:

|

Vulnerability

|

Severity

|

CVE

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CVE-2019-11254 — This is a Denial of Service (DoS) vulnerability that impacts the API server.

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Medium

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[CVE-2019-11254](#)

||

See the [GKE security bulletin](#) for instructions and more details.

GCP-2020-002

**Published: **2020-03-23 **Updated: **2020-03-23

Description

Kubernetes has disclosed the following vulnerabilities:

|
Vulnerability

|
Severity

|
CVE

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CVE-2020-8551 — This is a Denial of Service (DoS) vulnerability that impacts the kubelet.

|
Medium

|
[CVE-2020-8551](#)

|||

CVE-2020-8552 — This is a Denial of Service (DoS) vulnerability that impacts the API server.

|
Medium

|
[CVE-2020-8552](#)

||

See the [GKE security bulletin](#) for instructions and more details.

GCP-2020-001

****Published:** **2020-01-21 ****Updated:** **2020-01-21

Description

Microsoft has disclosed the following vulnerability:

|
Vulnerability

|
Severity

|

CVE

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CVE-2020-0601 — This vulnerability is also known as the **Windows Crypto API Spoofing Vulnerability**. It could be exploited to make malicious executables appear trusted or allow the attacker to conduct man-in-the-middle attacks and decrypt confidential information on user connections to the affected software.

|

NVD Base Score: 8.1 (High)

|

[CVE-2020-0601](#)

||

For more information, see the [Microsoft disclosure](#).

Google Cloud impact

The infrastructure hosting the Google Cloud and Google products is not impacted by this vulnerability. Additional per-product details are listed below.

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Product

|

Impact

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Compute Engine

|

[CVE-2020-0601](#)

For most customers, no further action is required.

Customers using Compute Engine virtual machines running Windows Server should ensure their instances have the latest Windows patch or use Windows Server images provided since 1/15/2020. Please see the [Compute Engine security bulletin](#) for more details.

|||

Google Kubernetes Engine

|

[CVE-2020-0601](#)

For most customers, no further action is required.

Customers using GKE with Windows Server nodes, both the nodes and the containerized workloads that run on those nodes must be updated to patched versions to mitigate this vulnerability. Please see the [GKE security bulletin](#) for instructions and more details.

|||

Managed Service for Microsoft Active Directory

|

[CVE-2020-0601](#)

For most customers, no further action is required.

All Managed Microsoft AD domains have been automatically updated with the patched image. Any customers manually running Microsoft Active Directory (and not utilizing Managed Microsoft AD) should ensure their instances have the latest Windows patch or use Windows Server images provided since 1/15/2020.

|||

Google Workspace

|

No customer action is required.

This service is not impacted by this vulnerability.

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App Engine standard environment

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No customer action is required.

This service is not impacted by this vulnerability.

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App Engine flexible environment

|

No customer action is required.

This service is not impacted by this vulnerability.

|||

Cloud Run

|

No customer action is required.

This service is not impacted by this vulnerability.

|||

Cloud Run functions

|

No customer action is required.

This service is not impacted by this vulnerability.

|||

Managed Service for Apache Airflow

|

No customer action is required.

This service is not impacted by this vulnerability.

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Dataflow

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No customer action is required.

This service is not impacted by this vulnerability.

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Managed Service for Apache Spark

|

No customer action is required.

This service is not impacted by this vulnerability.

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Cloud SQL

|

No customer action is required.

This service is not impacted by this vulnerability.

||

GCP-2019-001

Published: 2019-11-12 ****Updated:** **2019-11-12

Description

Intel has disclosed the following vulnerabilities:

|

Vulnerability

|

Severity

|

CVE

|||

CVE-2019-11135 — This vulnerability referred to as **TSX Async Abort (TAA)** can be used to exploit speculative execution within a **TSX transaction**. This vulnerability potentially allows data to be exposed via the same microarchitectural data structures exposed by [Microarchitectural Data Sampling \(MDS\)](#).

|
Medium

|
[CVE-2019-11135](#)

|||

CVE-2018-12207 — This is a **Denial of Service (DoS)** vulnerability affecting **virtual machine hosts (not guests)**. This issue is known as "Machine Check Error on Page Size Change."

|
Medium

|
[CVE-2018-12207](#)

||

For more information, see the Intel disclosures:

- [Intel Blog Post](#)
- [Intel's Developer Guidance Site:](#)
- [Machine Check Error on Page Size Change](#)
- [TSX Asynchronous Abort](#)

Google Cloud impact

The infrastructure hosting the Google Cloud and Google products is protected from these vulnerabilities. Additional per-product details are listed below.

|
Product

|
Impact

|||

Compute Engine

|
[CVE-2019-11135](#)

For most customers, no additional action is required.

N2, C2 or M2 customers running untrusted code in their own multi-tenant services within Compute Engine virtual machines should [stop and start their VMs](#) to ensure that they have the latest security mitigations.

[CVE-2018-12207](#)

For all customers, no additional action is required.

|||

Google Kubernetes Engine

|

[CVE-2019-11135](#)

For most customers, no additional action is required.

If you use node pools with N2, M2, or C2 nodes, and those nodes run untrusted code inside your own multi-tenant GKE clusters, then you should restart your nodes. If you want to restart all nodes in your node pool, [upgrade the affected node pool](#).

[CVE-2018-12207](#)

For all customers, no additional action is required.

|||

App Engine standard environment

|

No additional action is required.

|||

App Engine flexible environment

|

[CVE-2019-11135](#)

No additional action is required.

Customers should review Intel best practices with respect to application-level sharing which may occur between hyperthreads within a Flex VM.

[CVE-2018-12207](#)

No additional action is required.

|||

Cloud Run

|

No additional action is required.

|||

Cloud Run functions

|

No additional action is required.

|||

Managed Service for Apache Airflow

|

No additional action is required.

|||

Dataflow

|

[CVE-2019-11135](#)

For most customers, no additional action is required.

Dataflow customers who run multiple untrusted workloads on N2, C2, or M2 Compute Engine VMs managed by Dataflow and are concerned about intra-guest attacks should consider [restarting](#) any streaming pipelines that are currently running. Optionally, batch pipelines can be cancelled and re-run. No action is required for pipelines launched after today.

[CVE-2018-12207](#)

For all customers, no additional action is required.

|||

Managed Service for Apache Spark

|

[CVE-2019-11135](#)

For most customers, no additional action is required.

Cloud Dataproc customers who run multiple, untrusted workloads on the same Cloud Dataproc cluster running on Compute Engine N2, C2 or M2 VMs and are concerned about intra-guest attacks, should [redploy their clusters](#).

[CVE-2018-12207](#)

For all customers, no additional action is required.

|||

Cloud SQL

|

No additional action is required.

||

Send feedback

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